

Libertinus and diacritics

Anchor model

Libertinus has an anchor model where the base anchors are set to locations where the marks would be placed relative to the base. See the blue dots in the left panel of Figure 1; from top to bottom, the above-anchor is the blue dot above the base glyph, the overlay-anchor is blue dot inside the base glyph outline, and the below-anchor is the blue dot below the base glyph. The mark anchor is generally near the optical center of the mark glyph outline. See the red square in the middle panel of Figure 1; the mark's below-anchor is at the optical center of the mark. The base and mark are combined in the right panel of Figure 1 by superimposing their below-anchors.

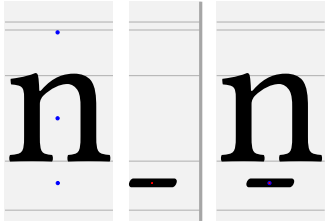


Figure 1: Libertinus n and macron with anchors.

Other fonts use a different geometric model for anchors. In Noto, the base's below-anchor is at the baseline, the overlay-anchor is inside the glyph, and the above-anchor is at the x-height. See the blue dots in the left panel of Figure 2; from the bottom to the top, the mark's below-anchor is the blue dot on the baseline, the overlay-anchor is blue dot inside the base glyph outline, and the above-anchor is the blue dot at the x-height. In Noto, the mark's below-anchor is at the origin (0, 0), and the mark glyph outline is drawn below the origin. In the middle panel of Figure 2, the mark's below-anchor is the red square at the origin. The base and mark are combined in the right panel of Figure 2 by superimposing their below-anchors.

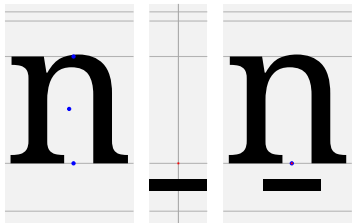


Figure 2: Noto n and macron with anchors.

Font metrics per glyph

See above- and below-anchor coordinates for A–Z, a–z, and IPA letters for Libertinus serif fonts (regular, italic, semibold, semibold italic) in four tables (Tables 1, 2, 3, 4, 5, 6). Recall that Language Science Press uses semibold Libertinus fonts, e.g. `LibertinusSerif-Semibold.otf`, for all boldface style.

Table row headings

hex: Unicode character codepoint in 4-hex.

reg, it, sb, si: regular, italic, semibold, semibold italic. In that row, the character is printed alone, with combining dot above, and with combining macron below, to show any anchor positioning. If no above- or below-anchor exists for that character, the character and combining mark are positioned by HarfBuzz fallback shaping.

ax, ay : X and Y coordinates of the above-anchor.

bx, by : X and Y coordinates of the below-anchor.

The next few rows use the bounding box of the glyph, because the midpoint of the width of the bounding box is a naive candidate for ax and bx , and the bounding box coordinates are easy to determine.

$\square xm$: midpoint of the bounding box X coordinates, $(X_{\min} + X_{\max})/2$, where the bounding box is characterized by $(X_{\min}, Y_{\min})$ and $(X_{\max}, Y_{\max})$.

$ax\delta$: X coordinate offset of above-anchor from midpoint of bounding box, $ax - \square xm$.

$ax\%$: normalized offset, $ax\delta$ divided by bounding box width, $ax\delta / (X_{\max} - X_{\min})$.

$bx\delta$: X coordinate offset of below-anchor from midpoint of bounding box, $bx - \square xm$.

$bx\%$: normalized offset, $bx\delta$ divided by bounding box width, $bx\delta / (X_{\max} - X_{\min})$.

There are better ways to define the geometric horizontal center of a glyph than the bounding box midpoint, depending on the outline at the top or bottom of the glyph.

Vertical aspects, anchor Y coordinates and reference, and clearance

A font has at least five vertical aspects where glyph outlines live inside: descender height, baseline, x-height, ascender height, capital height. The values or ranges of these Y coordinates are in the first five rows of Table 9. These vertical aspects are the gray horizontal lines of Figure 1. These vertical aspects are displayed in a FontForge glyph outline window as black horizontal guidelines with red or blue shaded bands for ranges. I denote these as $yra, yrc, yrx, yr0, yrd$ because they are Y values, and r indicates a reference value. a, c, x, 0, d indicate the five vertical aspects.

In the anchor model for Libertinus, the Y coordinate of an above-anchor (ay) or a below-anchor (by) is at a stable value depending on where the glyph outline lives relative to a vertical aspect. See the rows for ay and by in Tables 1 and 2 and find the common values according to the glyph's vertical aspects. There is no set name in font design for these above- or below-anchor Y coordinate reference values by ascender aspect, so I will denote them $ayra, ayrc, ayrx, byr0, byrd$. These are the middle five rows of Table 9. For example, all glyphs that sit on the baseline have below-anchor Y coordinate $by = -110$, and thus $byr0 = -110$. All capitals A–Z have $ay = 850$, and thus $ayrc = 850$. Any glyph that descends to the descender value $yrd = -232$ or descender range $[-238, -227]$ should have $by = byrd = -319$.

The clearance is defined as the difference between the anchor reference value and the vertical aspect Y value. I will denote these clearances as $ay\delta a, ay\delta c, ay\delta x, by\delta 0, by\delta d$. So, $ay\delta a = ayra - yra$ etc. For example, capital letters A–Z have a capital height $yrc = 645$, and all capital letters A–Z have an above-anchor Y coordinate at $ayrc = 850$. So the above-anchor clearance for capitals is $ay\delta c = ayrc - yec = 850 - 645 = 205$. Because of the weird anchor model for Libertinus, this clearance value is not the exact vertical distance between a base and a mark, but this clearance value is still useful.

A glyph may not be a full ascender or a full descender, and its bounding box may not live closely within two vertical aspects. In such a case, the anchor Y coordinate is set by looking for the nearest, or most suitable, vertical aspect, and considering the clearance of that vertical aspect relative to the bounding box of the glyph.

Here is the heuristic algorithm for calculating the Y coordinate of the above- or below-anchor (ay or by) of a glyph that is not a full ascender or descender.

Table 1: Font metrics for A–Z

hex	0041	0042	0043	0044	0045	0046	0047	0048	0049	004A	004B	004C	004D	004E	004F	0050	0051	0052	0053	0054	0055	0056	0057	0058	0059	005A
reg	AAA	BBB	CCC	DDD	EEE	FFF	GGG	HHH	III	JJJ	KKK	LLL	MMM	NNN	OOO	PPP	QQQ	RRR	SSS	TTT	UUU	VVV	WWW	XXX	YYY	ZZZ
ax	354	267	340	306	281	284	365	361	150	182	336	151	431	346	335	277	348	269	232	307	346	336	450	332	295	320
ay	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850
bx	308	268	363	309	287	209	367	362	147	109	342	262	393	348	350	240		258	234	300	328	318	479	307	295	287
by	-110	-110	-110	-110	-110	-110	-110	-110	-110	-319	-110	-110	-110	-110	-110	-110		-110	-110	-110	-110	-110	-110	-110	-110	-110
□xm	346	280	324	336	271	244	351	362	148	120	323	260	418	353	351	259	359	301	235	297	331	325	474	323	287	309
axδ	8	-13	16	-30	10	39	13	-1	2	62	13	-109	13	-7	-16	17	-11	-32	-3	10	15	10	-24	9	7	10
ax%	0.01	-0.03	0.03	-0.05	0.02	0.09	0.02	-0.00	0.01	0.16	0.02	-0.22	0.02	-0.01	-0.03	0.04	-0.02	-0.06	-0.01	0.02	0.02	0.02	-0.03	0.01	0.01	0.02
bxδ	-38	-12	39	-27	16	-35	15	0	-1	-11	19	1	-25	-5	-1	-19		-43	-1	3	-3	-7	5	-16	7	-22
bx%	-0.06	-0.02	0.07	-0.04	0.03	-0.08	0.02	0.00	-0.00	-0.03	0.03	0.00	-0.03	-0.01	-0.00	-0.04		-0.07	-0.00	0.01	-0.00	-0.01	0.01	-0.02	0.01	-0.04
it	AAA	BBB	CCC	DDD	EEE	FFF	GGG	HHH	III	JJJ	KKK	LLL	MMM	NNN	OOO	PPP	QQQ	RRR	SSS	TTT	UUU	VVV	WWW	XXX	YYY	ZZZ
ax	505	447	500	396	413	406	513	481	316	392	466	389	557	476	515	426	498	419	402	457	486	481	610	482	435	450
ay	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850
bx	308	238	333	279	269	131	345	320	127	109	292	252	363	328	310	140		248	204	277	308	263	439	287	270	267
by	-108	-108	-110	-107	-107	-111	-110	-107	-109	-319	-110	-110	-110	-109	-113	-110		-111	-111	-109	-109	-110	-107	-107	-109	-107
□xm	334	289	379	355	282	282	395	398	203	263	370	268	445	400	396	295	396	295	272	383	441	418	587	367	393	357
axδ	170	158	120	41	131	123	118	83	113	128	95	120	112	76	118	131	101	124	130	73	45	63	23	114	42	93
ax%	0.26	0.29	0.21	0.06	0.24	0.23	0.19	0.11	0.31	0.28	0.14	0.24	0.13	0.10	0.19	0.23	0.16	0.22	0.29	0.13	0.07	0.10	0.03	0.16	0.08	0.15
bxδ	-26	-51	-46	-76	-13	-151	-50	-78	-76	-154	-78	-16	-82	-72	-86	-155		-47	-68	-106	-133	-155	-148	-80	-123	-90
bx%	-0.04	-0.09	-0.08	-0.11	-0.02	-0.28	-0.08	-0.10	-0.21	-0.33	-0.11	-0.03	-0.10	-0.09	-0.14	-0.28		-0.08	-0.15	-0.20	-0.21	-0.25	-0.16	-0.11	-0.23	-0.14
sb	AAA	BBB	CCC	DDD	EEE	FFF	GGG	HHH	III	JJJ	KKK	LLL	MMM	NNN	OOO	PPP	QQQ	RRR	SSS	TTT	UUU	VVV	WWW	XXX	YYY	ZZZ
ax										203						363	363	304	252	298		331	493	364	316	321
ay										806						802	802	808	802	802		805	805	804	805	803
bx								362		109								292	234	301		303	470	318	316	308
by								-110		-319								-111	-111	-107		-108	-107	-107	-109	-107
□xm	349	299	332	351	283	268	368	362	160	131	340	267	427	354	365	277	365	320	237	299	339	318	489	354	308	309
axδ										72						-2	-2	-16	14	-1		12	4	9	7	12
ax%										0.17						-0.00	-0.00	-0.03	0.04	-0.00		0.02	0.00	0.01	0.01	0.02
bxδ							0			-22								-28	-3	2		-15	-19	-36	7	-1
bx%							0.00			-0.05								-0.05	-0.01	0.00		-0.02	-0.02	-0.05	0.01	-0.00
si	AAA	BBB	CCC	DDD	EEE	FFF	GGG	HHH	III	JJJ	KKK	LLL	MMM	NNN	OOO	PPP	QQQ	RRR	SSS	TTT	UUU	VVV	WWW	XXX	YYY	ZZZ
ax	354	267	340	306	273	276	365	361	150	432	336	149	431	346	335	316	348	269	232	307	346	336	450	332	295	320
ay	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850
bx	308	268	363	309	279	201	367	365	147	109	342	260	393	348	350	240		258	234	300	328	318	479	307	295	287
by	-110	-110	-110	-110	-110	-110	-110	-110	-110	-319	-110	-110	-110	-110	-110	-110		-110	-110	-110	-110	-110	-110	-110	-110	-110
□xm	364	298	378	367	305	304	405	445	244	288	421	312	449	452	435	307	435	309	263	381	470	417	581	387	407	350
axδ	-10	-31	-38	-61	-32	-28	-40	-84	-94	144	-85	-163	-18	-106	-100	8	-87	-40	-31	-74	-124	-81	-131	-55	-112	-30
ax%	-0.01	-0.05	-0.06	-0.09	-0.06	-0.05	-0.06	-0.11	-0.23	0.30	-0.12	-0.31	-0.02	-0.13	-0.15	0.01	-0.13	-0.07	-0.07	-0.13	-0.18	-0.13	-0.14	-0.07	-0.19	-0.05
bxδ	-56	-30	-15	-88	-26	-103	-38	-80	-97	-179	-79	-52	-56	-104	-85	-67		-51	-29	-81	-142	-99	-102	-80	-112	-63
bx%	-0.08	-0.05	-0.03	-0.13	-0.05	-0.18	-0.06	-0.10	-0.23	-0.37	-0.11	-0.10	-0.07	-0.13	-0.13	-0.12		-0.09	-0.07	-0.14	-0.21	-0.16	-0.11	-0.11	-0.19	-0.10

Table 2: Font metrics for a–z

hex	0061	0062	0063	0064	0065	0066	0067	0068	0069	006A	006B	006C	006D	006E	006F	0070	0071	0072	0073	0074	0075	0076	0077	0078	0079	007A
reg	<i>aȧ</i>	<i>bḃ</i>	<i>cċ</i>	<i>dḋ</i>	<i>eė</i>	<i>fḟ</i>	<i>gġ</i>	<i>hḣ</i>	<i>ii̇</i>	<i>jj̇</i>	<i>kk̇</i>	<i>ll̇</i>	<i>mṁ</i>	<i>nṅ</i>	<i>oȯ</i>	<i>pṗ</i>	<i>qq̇</i>	<i>rṙ</i>	<i>sṡ</i>	<i>tṫ</i>	<i>uu̇</i>	<i>vv̇</i>	<i>wẇ</i>	<i>xẋ</i>	<i>yẏ</i>	<i>zż</i>
ax	209		242	218	243		231	299	137		306	135	392	264	238	271	238	191	206		255	256	345	256	262	220
ay	645		645	645	645		645	645	795		784	885	645	646	645	646	645	646	645		645	646	646	646	645	646
bx	196	234	236	221	234	132	229	272	136	109	290	135	408	265	243	179	348	135	196	163	239	241	352	224	242	207
by	-110	-110	-110	-110	-110	-110	-319	-110	-110	-319	-214	-110	-110	-110	-110	-319	-319	-110	-110	-110	-110	-110	-110	-110	-319	-110
□xm	245	231	217	270	222	208	256	270	142	75	261	134	401	275	252	242	268	190	199	163	271	249	373	245	260	220
axδ	-36		24	-52	21		-25	28	-5		45	1	-9	-11	-14	28	-30	1	7		-16	7	-28	11	2	0
ax%	-0.09		0.07	-0.11	0.06		-0.06	0.06	-0.02		0.09	0.00	-0.01	-0.02	-0.03	0.06	-0.07	0.00	0.02		-0.03	0.01	-0.04	0.02	0.00	-0.00
bxδ	-49	2	18	-49	12	-76	-27	1	-6	33	29	1	6	-10	-9	-63	79	-55	-3	0	-32	-8	-21	-21	-18	-13
bx%	-0.12	0.01	0.05	-0.11	0.03	-0.21	-0.06	0.00	-0.03	0.14	0.06	0.00	0.01	-0.02	-0.02	-0.13	0.17	-0.16	-0.01	0.00	-0.06	-0.02	-0.03	-0.05	-0.04	-0.04
it	<i>aȧ</i>	<i>bḃ</i>	<i>cċ</i>	<i>dḋ</i>	<i>eė</i>	<i>fḟ</i>	<i>gġ</i>	<i>hḣ</i>	<i>ii̇</i>	<i>jj̇</i>	<i>kk̇</i>	<i>ll̇</i>	<i>mṁ</i>	<i>nṅ</i>	<i>oȯ</i>	<i>pṗ</i>	<i>qq̇</i>	<i>rṙ</i>	<i>sṡ</i>	<i>tṫ</i>	<i>uu̇</i>	<i>vv̇</i>	<i>wẇ</i>	<i>xẋ</i>	<i>yẏ</i>	<i>zż</i>
ax	341		343	218	332		360	409			405	292	522	394	359	411	369	312	327		386		348	366	367	360
ay	647		645	645	645		645	645			646	890	645	646	645	646	645	646	645		646		646	646	644	646
bx	174	214	163	221	158	222	197	229	136	0	223	115	382	235	186	248	169	113	169	133	226		355	224	368	187
by	-108	-108	-113	-107	-108	-111	-319	-111	-107	-319	-107	-107	-105	-111	-107	-110	-110	-110	-111	-107	-111		-110	-110	-107	-111
□xm	285	295	253	314	253	222	261	308	202	112	298	202	449	304	271	245	294	259	218	232	311	300	408	285	299	260
axδ	56		90	-96	78		98	100			107	90	73	89	87	166	75	53	109		75		-60	80	68	99
ax%	0.14		0.26	-0.21	0.23		0.19	0.23			0.24	0.41	0.11	0.21	0.23	0.31	0.18	0.16	0.34		0.17		-0.09	0.16	0.14	0.24
bxδ	-111	-81	-90	-93	-95	0	-64	-79	-66	-112	-75	-87	-67	-69	-85	3	-125	-146	-49	-99	-85		-53	-61	69	-73
bx%	-0.27	-0.21	-0.26	-0.20	-0.28	-0.00	-0.13	-0.18	-0.35	-0.29	-0.17	-0.40	-0.10	-0.16	-0.22	0.01	-0.29	-0.43	-0.15	-0.38	-0.19		-0.08	-0.12	0.14	-0.18
sb	<i>aȧ</i>	<i>bḃ</i>	<i>cċ</i>	<i>dḋ</i>	<i>eė</i>	<i>fḟ</i>	<i>gġ</i>	<i>hḣ</i>	<i>ii̇</i>	<i>jj̇</i>	<i>kk̇</i>	<i>ll̇</i>	<i>mṁ</i>	<i>nṅ</i>	<i>oȯ</i>	<i>pṗ</i>	<i>qq̇</i>	<i>rṙ</i>	<i>sṡ</i>	<i>tṫ</i>	<i>uu̇</i>	<i>vv̇</i>	<i>wẇ</i>	<i>xẋ</i>	<i>yẏ</i>	<i>zż</i>
ax						231			137			142	396	264							264					
ay						645			795			885	645	646							646					
bx						229	277	136	109			135	396	265						163	248					
by						-319	-110	-107	-319			-110	-105	-111						-107	-111					
□xm	255	235	224	273	232	225	263	275	155	78	261	148	408	289	259	254	273	212	201	168	275	262	384	260	264	221
axδ						-32		-18				-6	-12	-25							-11					
ax%						-0.07		-0.08				-0.03	-0.02	-0.05							-0.02					
bxδ						-34	2	-19	31			-13	-12	-24						-5	-27					
bx%						-0.08	0.00	-0.08	0.11			-0.05	-0.02	-0.05							-0.02	-0.05				
si	<i>aȧ</i>	<i>bḃ</i>	<i>cċ</i>	<i>dḋ</i>	<i>eė</i>	<i>fḟ</i>	<i>gġ</i>	<i>hḣ</i>	<i>ii̇</i>	<i>jj̇</i>	<i>kk̇</i>	<i>ll̇</i>	<i>mṁ</i>	<i>nṅ</i>	<i>oȯ</i>	<i>pṗ</i>	<i>qq̇</i>	<i>rṙ</i>	<i>sṡ</i>	<i>tṫ</i>	<i>uu̇</i>	<i>vv̇</i>	<i>wẇ</i>	<i>xẋ</i>	<i>yẏ</i>	<i>zż</i>
ax	379		242	218	243		291	469	137		306	174	392	264	238	271	238	191	326		255	256	345		262	220
ay	645		645	645	645		645	645	795		784	890	645	646	645	646	645	646	645		645	646	646		645	646
bx	196	234	236	221	234	132	199	272	136	67	290	135	400	265	243	179	348	135	196	133	239	241	352		242	207
by	-110	-110	-110	-110	-110	-110	-319	-110	-110	-319	-214	-110	-110	-110	-110	-319	-319	-110	-110	-107	-110	-110	-110		-319	-110
□xm	317	303	277	336	277	219	270	338	221	132	339	226	473	344	294	257	313	295	245	259	324	311	418	297	282	267
axδ	61		-35	-118	-34		21	130	-84		-33	-52	-81	-80	-56	13	-75	-104	80		-69	-55	-73		-20	-47
ax%	0.13		-0.09	-0.22	-0.09		0.04	0.26	-0.38		-0.07	-0.20	-0.11	-0.16	-0.13	0.02	-0.16	-0.25	0.22		-0.14	-0.13	-0.12		-0.04	-0.11
bxδ	-121	-69	-41	-115	-43	-87	-71	-66	-85	-65	-49	-91	-73	-79	-51	-78	35	-160	-49	-126	-85	-70	-66		-40	-60
bx%	-0.25	-0.14	-0.10	-0.22	-0.11	-0.14	-0.13	-0.13	-0.38	-0.15	-0.10	-0.35	-0.10	-0.16	-0.12	-0.13	0.07	-0.39	-0.14	-0.40	-0.17	-0.16	-0.11		-0.08	-0.14

hex	00C6	00D0	00D8	00DE	00DF	00E6	00F0	00F8	00FE	0110	0111	0126	0127	0131	0141	0142	0152	0153	017F	0181	0182	0186	0187	0189	018A	018B	018E	018F	0190	0191	0192	0193	
reg	ÆĖĖ	ÐÐÐ	ÐÐÐ	ÐÐÐ	ÐÐÐ	æææ	ððð	øøø	þþþ	ÐÐÐ	ddd	HHH	hhh	ii	LLL	lll	ĖĖĖ	æææ	ff	B̢B̢	B̢B̢	ƵƵƵ	CCC	ÐÐÐ	ÐÐÐ	ÐÐÐ	ÐÐÐ	ÐÐÐ	ÐÐÐ	ÐÐÐ	ÐÐÐ	ÐÐÐ	ÐÐÐ
ax	480	341	351	281		350			278					132			460	405		327					306								
ay	850	850	850	850		645			646					645			850	645		805					805								
bx	549		350			327		243						139						328					369								
by	-107		-110			-109		-110						-108						-108					-107								
□xm	426	333	351	249	259	343	240	250	236	336	270	362	270	142	256	133	436	387	190	303	273	322	369	333	368	292	280	326	256	183	201	371	
axδ	53	8	0	31		6			41					-10			24	17		24					-62								
ax%	0.06	0.01	-0.00	0.06		0.01			0.09					-0.04			0.03	0.02		0.04					-0.09								
bxδ	122		-1			-16			-7					-3						25					0								
bx%	0.14		-0.00			-0.03		-0.02						-0.01						0.04					0.00								
it	ÆĖĖ	ÐÐÐ	ÐÐÐ	ÐÐÐ	ÐÐÐ	æææ	ððð	øøø	þþþ	ÐÐÐ	ddd	HHH	hhh	ii	LLL	lll	ĖĖĖ	æææ	ff	B̢B̢	B̢B̢	ƵƵƵ	CCC	ÐÐÐ	ÐÐÐ	ÐÐÐ	ÐÐÐ	ÐÐÐ	ÐÐÐ	ÐÐÐ	ÐÐÐ	ÐÐÐ	ÐÐÐ
ax	615					480			278					267			575	480		327					306								
ay	850					645			646					645			850	645		805					805								
bx	549		310			327		186						139						328					369								
by	-107		-113			-109		-107						-108						-108					-107								
□xm	433	355	411	274	244	394	298	283	247	355	333	398	308	197	268	199	481	399	217	347	314	386	483	375	433	358	377	411	307	228	268	462	
axδ	181					85			30					70			94	81		-20					-127								
ax%	0.21					0.13			0.06					0.39			0.12	0.13		-0.04					-0.19								
bxδ	115		-101			-67		-97						-58						-19					-64								
bx%	0.13		-0.15			-0.10		-0.22						-0.32						-0.03					-0.09								

Table 4: Font metrics for Latin and IPA 2/4

hex	0194	0196	0197	0198	019C	019D	019F	01B7	0237	0250	0251	0252	0253	0254	0255	0256	0257	0258	0259	025A	025B	025C	025D	025E	025F	0260	0261	0262	0263	0264	0265	0266
reg	ŸŸŸ	lll	lll	KKK	llllll	NNN	000	333	jjj	eee	aaa	ooo	bbb	ddd	eee	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd
ax	331				431				140	228	228	264	480	216	209	218	218	204	195	229	209	194	209	239	169	209	239	265	259	248	268	299
ay	805				801				645	645	645	645	630	645	645	645	645	645	645	645	645	645	645	645	645	645	645	645	645	645	645	565
bx					450		350	272		229	212	259	212	177	212	212	212	212	186	212	212	184	212	256	196	236	226	248	251	216	219	240
by					-108		-110	-319		-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-110	-110	-320	-110	-277	-114	-129	-114
□xm	308	151	148	339	433	324	351	303	74	216	272	247	265	215	222	314	322	220	232	306	201	195	234	241	124	320	228	259	258	221	271	262
axδ	22				-2				66	11	-44	17	215	0	-13	-96	-104	-16	-37	-77	7	-1	-25	-2	44	-111	11	6	1	27	-3	36
ax%	0.04				-0.00				0.27	0.03	-0.09	0.04	0.56	0.00	-0.04	-0.17	-0.18	-0.04	-0.10	-0.14	0.02	-0.00	-0.06	-0.01	0.13	-0.20	0.03	0.01	0.00	0.06	-0.01	0.07
bxδ					16		-1	-31		12	-60	12	-53	-38	-10	-102	-110	-8	-46	-94	10	-11	-22	14	71	-84	-2	-11	-7	-5	-52	-22
bx%					0.02		-0.00	-0.06		0.03	-0.13	0.03	-0.14	-0.11	-0.03	-0.19	-0.19	-0.02	-0.12	-0.17	0.03	-0.04	-0.06	0.03	0.21	-0.15	-0.01	-0.02	-0.01	-0.01	-0.10	-0.05
it	ŸŸŸ	lll	lll	KKK	llllll	NNN	000	333	jjj	eee	aaa	ooo	bbb	ddd	eee	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd
ax	331				431				140					322		218	218	204			322						265			258	299	
ay	805				801				645					654		645	645	645			654						645			645	565	
bx					450					229	212	243	212	172	212	212	212	212	212	212	212	184	212	256			248		177	219	240	
by					-108					-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114			-114		-114	-129	-114	
□xm	392	202	203	375	536	368	405	341	96	299	297	303	286	236	253	325	396	260	268	372	252	228	315	292	157	402	279	291	346	297	313	309
axδ	-61				-105				43					86		-107	-178	-56			69						-26			-55	-10	
ax%	-0.10				-0.13				0.12					0.25		-0.21	-0.28	-0.16			0.19						-0.06			-0.13	-0.02	
bxδ					-86					-70	-85	-60	-74	-64	-41	-113	-184	-48	-56	-160	-40	-44	-103	-36			-43		-120	-94	-69	
bx%					-0.10					-0.17	-0.19	-0.14	-0.17	-0.19	-0.11	-0.23	-0.29	-0.14	-0.16	-0.27	-0.11	-0.12	-0.19	-0.08			-0.10		-0.34	-0.22	-0.16	
sb	ŸŸŸ	lll	lll	KKK	llllll	NNN	000	333	jjj	eee	aaa	ooo	bbb	ddd	eee	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd
ax									163					199							229											
ay									645					645							645											
bx																																
by									272		228			184					212													
□xm	293	169	158	337	444	336	365	315	78	207	276	251	268	199	232	322	323	230	228	319	212	215	276	249	144	336	235	265	272	202	277	274
axδ									85					0							16											
ax%									0.30					0.00							0.05											
bxδ									-43		21			-15					-16													
bx%									-0.08		0.05			-0.04					-0.04													
si	ŸŸŸ	lll	lll	KKK	llllll	NNN	000	333	jjj	eee	aaa	ooo	bbb	ddd	eee	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd	ddd
ax	331				431				140		228	264		160		218	218	204			209	194					265			258	299	
ay	805				801				645		645	645		645		645	645	645			645	645					645			645	565	
bx					450		236		229	212	259	212	177	212	212	212	212	212		212	212	184	212	256			248		177	219	240	
by					-108		-319		-114	-114	-114	-114	-114	-114	-114	-114	-114	-114		-114	-114	-114	-114	-114			-110		-114	-129	-114	
□xm	405	226	227	388	547	380	436	341	135	290	310	317	297	237	292	339	416	266	278	383	265	250	334	308	164	415	288	311	369	298	369	277
axδ	-74				-116				5		-82	-53		-77		-121	-198	-62			-56	-56					-46			-111	21	
ax%	-0.11				-0.13				0.01		-0.17	-0.11		-0.19		-0.22	-0.28	-0.14			-0.14	-0.14					-0.10			-0.21	0.04	
bxδ					-97		-105		-61	-98	-58	-85	-60	-80	-127	-204	-54			-171	-53	-66	-122	-52			-63		-121	-150	-37	
bx%					-0.11		-0.16		-0.14	-0.20	-0.12	-0.18	-0.15	-0.18	-0.24	-0.29	-0.12			-0.27	-0.13	-0.17	-0.22	-0.11			-0.13		-0.31	-0.29	-0.07	

Table 5: Font metrics for Latin and IPA 3/4

hex	0267	0268	0269	026A	026B	026C	026D	026E	026F	0270	0271	0272	0273	0274	0275	0277	0278	0279	027A	027B	027C	027D	027E	027F	0280	0281	0282	0283	0284	0285	0286	0287
reg	ḡḡḡ	ḡḡḡ	ḡḡ	ḡḡ	ḡḡ	ḡḡ	ḡḡ	ḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	
ax	299	155	123	133	209	209	209	209	407	209	383	264	263	209	209	209	209	209	209	209	222	209	209	209	209	209	209	209	209	209	209	209
ay	565	795	645	645	645	645	645	645	645	645	645	646	645	645	645	645	645	645	645	645	703	645	645	645	645	645	645	645	645	645	645	645
bx	240	134	153	139	146	218	196	196	410	294	383	265	262	247	196	341	196	190	203	196	127	196	196	196	196	196	196	196	196	196	196	196
by	-114	-114	-114	-114	-114	-120	-110	-110	-114	-114	-105	-111	-112	-114	-110	-114	-110	-114	-114	-110	-305	-110	-110	-110	-110	-110	-110	-110	-110	-110	-110	-110
□xm	227	149	170	136	144	182	182	250	390	379	365	234	295	278	252	357	285	177	179	225	186	186	174	170	238	238	199	140	140	170	149	150
axδ	71	5	-47	-3	65	27	27	-41	16	-170	17	29	-32	-69	-43	-148	-76	31	29		35	22	34	38	-29	-29	10	68	68	38	59	59
ax%	0.17	0.02	-0.23	-0.01	0.23	0.08	0.09	-0.09	0.02	-0.23	0.03	0.05	-0.06	-0.14	-0.10	-0.23	-0.15	0.09	0.09		0.11	0.07	0.11	0.12	-0.07	-0.07	0.03	0.17	0.17	0.09	0.15	0.21
bxδ	12	-15	-17	3	2	36	14	-54	19	-85	17	30	-33	-31	-56	-16	-89	12	23	-29	-59	9	21	25	-42	-42	-3	55	55	25	46	46
bx%	0.03	-0.06	-0.08	0.01	0.01	0.11	0.04	-0.12	0.03	-0.12	0.03	0.06	-0.06	-0.06	-0.13	-0.02	-0.18	0.04	0.07	-0.07	-0.19	0.03	0.07	0.08	-0.10	-0.10	-0.01	0.14	0.14	0.06	0.12	0.17
it	ḡḡḡ	ḡḡḡ	ḡḡ	ḡḡ	ḡḡ	ḡḡ	ḡḡ	ḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	
ax									407		383	264	263								222											
ay									645		645	646	645								703											
bx		134	153	139	146	218			410	294	383	265	262	247		341		190	203		127											
by		-114	-114	-114	-114	-120			-114	-114	-105	-111	-112	-114		-114		-114	-114		-305											
□xm	297	204	178	186	210	240	193	284	472	473	388	210	295	328	289	398	342	98	268	233	217	265	218	215	245	260	217	197	197	198	206	170
axδ			-55	-58					-65		-5	53	-32								5											
ax%			-0.31	-0.18					-0.09		-0.01	0.09	-0.08								0.01											
bxδ		-70	-25	-47	-64	-22			-62	-179	-5	54	-33	-81		-57		92	-65		-90											
bx%		-0.28	-0.14	-0.15	-0.22	-0.07			-0.09	-0.25	-0.01	0.09	-0.08	-0.14		-0.09		0.27	-0.13		-0.19											
sb	ḡḡḡ	ḡḡḡ	ḡḡ	ḡḡ	ḡḡ	ḡḡ	ḡḡ	ḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	
ax									377																							
ay									645																							
bx		134		156	146	218			380						229			190														
by		-114		-114	-114	-120			-114						-114			-114														
□xm	243	152	164	152	156	190	193	273	354	388	363	245	306	290	259	369	297	144	205	237	205	198	186	182	235	220	201	152	152	152	161	157
axδ				-6					23																							
ax%				-0.02					0.03																							
bxδ		-18		4	-10	27			26						-30			45														
bx%		-0.07		0.02	-0.04	0.08			0.03						-0.07			0.13														
si	ḡḡḡ	ḡḡḡ	ḡḡ	ḡḡ	ḡḡ	ḡḡ	ḡḡ	ḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	ḡḡḡḡ	
ax	299	137	123	128					407		383	264	263								222											
ay	565	795	645	645					645		645	646	645								703											
bx	240	136	153	139	146	218			410	294	383	265	262	247		341		190	203		127											
by	-114	-110	-114	-114	-114	-120			-114	-114	-105	-111	-112	-114		-114		-114	-114		-305											
□xm	279	208	215	198	227	288	235	298	467	484	401	227	321	340	321	412	347	225	280	246	230	272	230	228	257	274	244	209	209	208	215	162
axδ	19	-71	-92	-70					-60		-18	37	-58								-8											
ax%	0.04	-0.27	-0.39	-0.20					-0.08		-0.02	0.06	-0.09								-0.02											
bxδ	-39	-72	-62	-59	-81	-70			-57	-190	-18	38	-59	-93		-71		-35	-77		-103											
bx%	-0.08	-0.28	-0.26	-0.17	-0.23	-0.20			-0.08	-0.25	-0.02	0.06	-0.10	-0.15		-0.10		-0.08	-0.15		-0.20											

Table 6: Font metrics for Latin and IPA 4/4

hex	0288	0289	028A	028B	028C	028D	028E	028F	0290	0291	0292	0293	0294	0295	0296	0297	0299	029B	029C	029D	029F
reg	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt
ax	209	209	209	259	209	209	209	209	209	209	218	209	209	209	209	209	209	209	209	209	209
ay	645	645	645	645	645	645	645	645	645	645	646	645	645	645	645	645	645	645	645	645	645
bx	196	196	258	251	196	196	196	239	196	196	185	196	202	230	196	196	196	196	305	196	196
by	-110	-110	-114	-114	-110	-110	-110	-107	-110	-110	-319	-110	-107	-107	-110	-110	-110	-110	-110	-110	-110
□xm	157	271	253	235	244	367	248	236	262	217	216	220	211	221	208	221	232	294	293	126	207
axδ	51	-62	-44	24	-35	-158	-39	-27	-53	-8	1	-11	-2	-12	1	-12	-23	-85	-84	83	2
ax%	0.19	-0.12	-0.09	0.06	-0.07	-0.22	-0.08	-0.06	-0.12	-0.02	0.00	-0.03	-0.01	-0.03	0.00	-0.03	-0.06	-0.16	-0.16	0.29	0.01
bxδ	38	-75	4	16	-48	-171	-52	3	-66	-21	-31	-24	-9	9	-12	-25	-36	-98	12	70	-11
bx%	0.14	-0.15	0.01	0.04	-0.10	-0.23	-0.10	0.01	-0.15	-0.06	-0.09	-0.07	-0.02	0.02	-0.03	-0.07	-0.09	-0.19	0.02	0.24	-0.03
it	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt
ax				259							328										
ay				645							646										
bx			258	251				239			130		202	230					273		
by			-114	-114				-107			-319		-107	-107					-110		
□xm	224	318	312	323	248	372	305	332	251	251	236	246	286	293	242	290	245	361	330	145	222
axδ				-64							92										
ax%				-0.15							0.22										
bxδ			-54	-72				-93			-106		-84	-63					-57		
bx%			-0.11	-0.17				-0.21			-0.25		-0.21	-0.16					-0.10		
sb	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt
ax			266								234										
ay			645								645										
bx											193								351		
by											-319								-110		
□xm	182	274	267	255	231	355	243	244	254	229	227	236	206	206	206	273	245	290	316	132	219
axδ			-1								7										
ax%			-0.00								0.02										
bxδ											-34								35		
bx%											-0.08								0.06		
si	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt	ttt
ax				259							218										
ay				645							646										
bx			258	251				239			141		202	230					327		
by			-114	-114				-107			-319		-107	-107					-110		
□xm	237	349	325	336	241	367	303	341	281	264	237	259	297	306	255	303	267	390	404	159	235
axδ				-77							-19										
ax%				-0.17							-0.04										
bxδ			-67	-85				-102			-96		-95	-76					-77		
bx%			-0.12	-0.19				-0.22			-0.19		-0.21	-0.18					-0.11		

Table 7: Font metrics for above marks

hex	0300	0301	0302	0303	0304	0306	0307	0308	0309	030A	030B	030C	030D	030F	0310	0311	0312	033D	033E	033F	030E	0346	034A	034B	034C	0350	0351	0352	0357	035B	
reg	`	´	^	~	_	˘	˙	˚	˛	°	¨	˘	˙	˚	˛	°	ˆ	˜	˘	˙	˚	˛	˜	˘	˙	˚	˛	˜	˘	˙	˚
ax	-142	-191	-139	-149	-209	-161	-180	-210	-160	-184	-219	-205	-155	-140	-251	-205	-110	-265	-265	-235	-245	-251	-215	-205	-235	-201	-171	-251	-171	-164	
ay	683	682	697	697	644	681	667	645	765	708	714	676	688	758	675	645	605	605	605	611	719	621	645	645	645	645	645	645	645	645	
◻xm	-159	-160	-140	-149	-212	-161	-180	-210	-161	-182	-170	-204	-154	-168	-251	-208	-102	-261	-247	-240	-245	-255	-217	-217	-239	-183	-156	-251	-141	-175	
axδ	17	-31	1	0	2	0	0	0	1	-2	-49	-1	-1	27	0	3	-8	-4	-18	5	0	4	2	12	3	-18	-15	0	-30	10	
ax%	0.11	-0.20	0.00	0.00	0.01	0.00	0.00	-0.00	0.01	-0.01	-0.19	-0.01	-0.02	0.11	0.00	0.01	-0.08	-0.02	-0.21	0.01	-0.00	0.02	0.01	0.04	0.01	-0.08	-0.12	0.00	-0.23	0.05	
it	`	´	^	~	_	˘	˙	˚	˛	°	¨	˘	˙	˚	˛	°	ˆ	˜	˘	˙	˚	˛	˜	˘	˙	˚	˛	˜	˘	˙	˚
ax	-3	-52	3	-17	-78	-22	-40	-79	-4	-40	-74	-67	-15	14	-114	-74	13	-142	-142	-111	-99	-125	-84	-74	-104	-70	-40	-120	-40	-33	
ay	683	682	697	697	645	681	689	645	765	708	714	676	688	758	675	645	605	605	605	611	719	621	645	645	645	645	645	645	645	645	
◻xm	-33	-34	-24	-23	-98	-26	-55	-94	-14	-49	-41	-69	-26	-35	-116	-104	16	-151	-134	-129	-117	-146	-99	-97	-119	-71	-31	-131	-25	-52	
axδ	29	-18	27	5	19	4	15	15	9	8	-33	1	11	48	2	30	-3	8	-8	18	17	20	14	22	15	0	-9	11	-15	18	
ax%	0.22	-0.10	0.11	0.02	0.07	0.01	0.16	0.06	0.06	0.05	-0.12	0.01	0.13	0.20	0.01	0.11	-0.03	0.04	-0.08	0.04	0.09	0.08	0.05	0.08	0.05	0.00	-0.07	0.04	-0.11	0.09	
sb	`	´	^	~	_	˘	˙	˚	˛	°	¨	˘	˙	˚	˛	°	ˆ	˜	˘	˙	˚	˛	˜	˘	˙	˚	˛	˜	˘	˙	˚
ax	-127	-180	-127	-133	-205	-149					-197	-193	-137	-127	-239	-205					-232									-180	
ay	713	713	700	679	625	684					733	676	719	758	675	645					719									639	
◻xm	-145	-144	-128	-137	-200	-149	-168	-186	-149	-170	-158	-193	-137	-153	-235	-220	-102				-232					105		111	-184		
axδ	17	-36	0	3	-5	0					-39	0	0	25	-4	15					0									4	
ax%	0.10	-0.21	0.00	0.01	-0.02	0.00					-0.15	-0.00	0.00	0.10	-0.01	0.05					0.00									0.02	
si	`	´	^	~	_	˘	˙	˚	˛	°	¨	˘	˙	˚	˛	°	ˆ	˜	˘	˙	˚	˛	˜	˘	˙	˚	˛	˜	˘	˙	˚
ax	-142	-191	-139	-134	-209	-161	-180	-210	-160	-184	-219	-205	-155	-140	-251	-205	-110				-245										
ay	683	682	697	700	645	681	667	645	765	708	714	676	688	758	675	645	605				719										
◻xm	-18	-19	-12	-11	-92	-15	-45	-112	-1	-37	-30	-59	-9	-22	-101	-103	26				-105										
axδ	-124	-172	-127	-123	-117	-146	-135	-98	-159	-147	-189	-146	-146	-118	-150	-102	-136				-140										
ax%	-0.89	-0.92	-0.50	-0.42	-0.38	-0.54	-1.14	-0.34	-1.07	-0.84	-0.66	-0.57	-1.59	-0.48	-0.56	-0.38	-1.03				-0.76										

Table 8: Font metrics for below marks

hex	0316	0317	0318	0319	031C	0320	0323	0324	0325	0326	0329	032A	032B	032C	032D	032E	032F	031D	031E	031F	0330	0331	0333	0339	033A	033B	033C	0347	0348	0349	034D	034E
reg																																
bx	-219	-210	-224	-285	-216	-253	-216	-218	-218	-255	-257	-253	-251	-204	-195	-251	-250	-249	-247	-249	-213	-244	-244	-216	-253	-246	-251	-244	-197	-192	-222	-227
by	-133	-132	-148	-147	-146	-148	-113	-116	-116	-82	-122	-113	-108	-116	-133	-110	-116	-143	-106	-145	-70	-110	-110	-146	-113	-113	-108	-110	-110	-110	-110	-110
□xm	-248	-225	-249	-249	-262	-252	-216	-217	-217	-245	-256	-255	-252	-204	-195	-252	-251	-249	-249	-252	-205	-240	-240	-178	-255	-252	-252	-240	-195	-189	-229	-229
bxδ	29	14	25	-36	46	-1	0	-1	-1	-10	-1	2	0	0	0	1	1	0	2	2	-8	-4	-4	-38	2	6	0	-4	-2	-3	6	1
bx%	0.16	0.08	0.15	-0.22	0.42	-0.01	0.00	-0.00	-0.01	-0.09	-0.02	0.01	0.00	-0.00	0.00	0.00	0.00	0.00	0.01	0.01	-0.03	-0.02	-0.02	-0.35	0.01	0.03	0.00	-0.01	-0.01	-0.02	0.02	0.01
it																																
bx	-246	-237	-254	-315	-246	-283	-239	-242	-242	-272	-282	-276	-273	-228	-222	-275	-274	-278	-269	-278	-227	-266	-266	-246	-276	-269	-273	-266	-219	-214	-244	-249
by	-133	-132	-148	-147	-146	-148	-113	-116	-116	-82	-122	-113	-108	-116	-133	-110	-116	-143	-106	-145	-70	-110	-110	-146	-113	-113	-108	-110	-110	-110	-110	-110
□xm	-282	-257	-270	-287	-284	-281	-239	-240	-241	-278	-286	-286	-271	-222	-238	-261	-295	-291	-267	-281	-220	-262	-269	-217	-285	-284	-293	-269	-223	-206	-256	-259
bxδ	35	20	16	-28	37	-2	0	-2	-1	5	4	10	-2	-6	16	-14	20	13	-2	3	-7	-4	3	-29	8	14	19	3	4	-8	11	10
bx%	0.21	0.11	0.09	-0.16	0.29	-0.01	0.00	-0.01	-0.01	0.04	0.05	0.04	-0.01	-0.03	0.06	-0.05	0.08	0.07	-0.01	0.02	-0.02	-0.02	0.01	-0.23	0.03	0.07	0.06	0.01	0.02	-0.05	0.03	0.05
sb																																
bx	-219	-210	-224	-285	-216	-253	-216	-218	-218	-255	-257	-253	-251	-208	-195	-239	-250	-249	-247	-249	-213	-244										
by	-133	-132	-148	-147	-146	-149	-113	-116	-116	-73	-124	-113	-108	-116	-133	-110	-116	-143	-106	-146	-70	-58										
□xm	-248	-225	-249	-250	-262	-251	-217	-218	-217	-245	-256	-255	-252	-204	-195	-240	-263	-249	-250	-251	-193	-240										
bxδ	28	14	24	-35	46	-2	0	0	-1	-10	-1	2	0	-4	0	1	13	0	2	2	-20	-4										
bx%	0.16	0.08	0.15	-0.22	0.42	-0.01	0.00	-0.00	-0.01	-0.09	-0.02	0.01	0.00	-0.02	0.00	0.00	0.05	-0.00	0.01	0.01	-0.07	-0.01										
si																																
bx	-219	-210	-224	-285	-216	-253	-216	-218	-218	-255	-257	-253	-251	-204	-195	-251	-250	-249	-247	-249	-213	-244										
by	-133	-132	-148	-147	-146	-148	-113	-116	-116	-82	-122	-113	-108	-116	-133	-110	-116	-143	-106	-145	-70	-110										
□xm	-270	-245	-258	-288	-271	-269	-227	-228	-228	-261	-275	-274	-259	-211	-226	-249	-294	-281	-252	-269	-196	-239										
bxδ	50	35	34	2	55	16	11	10	10	6	17	21	7	7	31	-2	43	32	5	20	-17	-5										
bx%	0.30	0.19	0.19	0.01	0.42	0.08	0.09	0.03	0.06	0.04	0.20	0.08	0.02	0.03	0.12	-0.01	0.16	0.18	0.03	0.10	-0.06	-0.02										

Consider a glyph that is not a full ascender. Find the nearest, or most suitable, vertical aspect – ascender a , capital c , or x-height x . Then $ay = Y_{\max} + ay\delta[acx] - \epsilon$, where $Y_{\max}$ is the maximum Y coordinate of bounding box of the glyph, and ϵ is the correction for the vertical aspect range, usually for an overshoot.

Consider a glyph that is not a full descender. Find the nearest, or most suitable, vertical aspect – baseline 0 or descender d . Then $by = Y_{\min} + by\delta[0d] + \epsilon$, where $Y_{\min}$ is the minimum Y coordinate of the bounding box of the glyph, and ϵ is the correction for the vertical aspect range, usually for an overshoot.

Table 9: Vertical aspects, anchor Y references, and clearances of Libertinus

	v. aspect	regular	italic	semibold	semibold italic
<i>yra</i>	ascender	688, [688, 698]	688, [688, 698]	690	698
<i>ycr</i>	capital	645, [645, 658]	645, [645, 658]	645	645
<i>yrx</i>	x-height	429, [429, 432]	429, [429, 432]	433	434, [433, 447]
<i>yr0</i>	baseline	0, [-12, 0]	0, [-12, 0]	0	0
<i>yrd</i>	descender	-232, [-238, -227]	-233, [-238, -227]	-238	-232, [-238, -232]
<i>ayra</i>	ascender	885	890	885	890
<i>ayrc</i>	capital	850	850	805	850
<i>ayrx</i>	x-height	645	645	645	645
<i>byr0</i>	baseline	-110	-110	-110	-110
<i>byrd</i>	descender	-319	-319	-319	-319
<i>ayda</i>	ascender	187	202	195	192
<i>aydc</i>	capital	205	205	205	205
<i>aydx</i>	x-height	216	216	212	198
<i>byd0</i>	baseline	110	110	110	110
<i>bydd</i>	descender	87	87	81	100

Base coverage

Libertinus serif regular has good above- and below-anchor base coverage for A–Z, a–z, and IPA letters, in the sense that most anchors are defined. But I will show later that some anchors are not set to good values. The base coverage gets worse for italic and semibold. Observe that many IPA letters reuse $ax = 209$ and $bx = 196$. These are good anchors for the letter a, but these were just copied to some IPA letters without regard to the letterform and are therefore bad anchors for those IPA letters.

Anchor survey, with patches

I will systematically survey above- and below anchors in the four font styles, commenting on any features and patching any problems. Refer to Tables 1 and 2 for anchor values. Missing anchors are readily apparent as blank entries. Large anchor-midpoint offset values ($ax\delta$, $bx\delta$) indicate possible problems, but these can often be explained by glyph symmetry.

A–Z, regular, above- and below-anchor

The above-anchors of A–Z regular are good.

The above-anchor of D is just right of the major stem, not right enough to be centered over the bowl. For D, $ax = 306$. Consider the precomposed characters $\ddot{D}$ (U+010E) and $\dot{D}$ (U+1E0A). The designer drew their glyphs with the mark centered at $X = 312$ and 296 respectively. So, this is a deliberate design decision to keep the mark closer to the stem, not centered over the bowl.

Similarly, the above anchor of F is just right of the major stem, not right enough to be centered over cross strokes. For F, $ax = 284$. For $\dot{\text{F}}$ (U+1E1E), the dot is centered at $X = 254$. So, the designer is deliberately choosing to keep the mark closer to the stem.

The above-anchor of L is centered over the major stem, which is a valid design choice.

The below-anchors of A–Z regular are good. But J's below anchor is too low, and Q is missing a below-anchor. Even though J and Q arguably do not need below-anchors for linguistic notation, I will set this anchor by the heuristic algorithm.

Patch J regular, below-anchor

J's below-anchor is too low. For J, $by = 319$, but J is not a full descender; its $\text{bbox } Y_{\min} = -172$, which is well above $yr\text{d} = -232$. Yes, the designer set $by = byr\text{d} = -319$ for J in all four font styles, but this is arguably too low. According to the heuristic algorithm, the below-anchor anchor should be at $by = Y_{\min} - by\delta d + \epsilon = -172 - 87 + 5 = -254$, where $\epsilon = 5$ accounts for the overshoot of J's tail.

Patch Q regular, below-anchor

Q has $\text{bbox } Y_{\min} = -209$. Again, according to the heuristic algorithm, the below-anchor anchor should be at $by = Y_{\min} - by\delta d + \epsilon = -209 - 87 + 5 = -291$, where $\epsilon = 5$ accounts for the overshoot of Q's tail. For bx , observe that O and Q have the same counter outline, literally. So, the below-anchor for Q should be at $bx = 350$, the same as that for O.

a–z, regular, above- and below-anchor

b, f, j, and t are missing above-anchors. Some letters (n, p, r, v, w, x, z) should have ay regularized to 645, not 646. For d, h, and k, the above-anchor is centered above the body, not the ascender; this is good design choice, but care must be taken to avoid collision between an above-mark and an ascender. For d and h, ay is at the x-height anchor reference, and therefore wide marks collide with the ascender. For k, ay is a bit higher, but not at the full ascender anchor references, and therefore the widest marks collide with the ascender.

The below-anchor of k is too low. Otherwise, the below-anchors of a–z are good.

Patch k regular, below-anchor

For k, $by = 214$. But it should be at $byr0 = -110$, like all other letters that sit on the baseline.

Here is this positioning of k in a patched version of the font (with o for comparison only).

Original: $\text{k } \text{ko } \text{ok}$

Patched: $\text{k } \text{ko } \text{ok}$

Patch b, d, f, h regular, above-anchor

For b, d, f, and h, set $ay = ayr0 = 885$.

For ax for b, d, f, and h, there are at least two strategies. First, set ax at the bowl apex. So, for f, set $ax = 279$ at the apex of the bowl. But for b, d, and h, you could set ax centered horizontally over the bowl, in which case you also need to decide which two vertical aspects of the bowl should be used for the centering.

Patch j regular, above-anchor

j needs an above anchor for the rare case of an above-mark that is placed over the dot. i does have an above-anchor at (137, 795), and the outlines of the top of

Patch n, p, r, v, w, x, z regular, above-anchor

In my patched version of the font, I regularize *ay* to *ayr0* = 645 for all letters that sit at the x-height, including n, p, r, v, w, x, z.

Above-anchor, a–z regular

b, f, j, and t are missing the above-anchor. d and h have an above-anchor above the meanline but not above the ascender. l has an above-anchor above the ascender. k has an above-anchor, but it is just below the ascender, so any marks other than the do crash into the ascender. These are all legitimate design choices. There are no above-marks, other than those with precomposed characters, the would ever be combined with b, f, or t.

Patch j dot removal for three above-marks

The dot of i and j should be removed before adding any above-mark. But the dot is not removed for three above-marks: x, turned tilde, double macron. So the GSUB table should be patched to fir the dotless j substitution in this context.

To do.

Capital mark alternates

Libertinus has custom glyphs for some marks (acute, grave, circumflex, breve, a.o.) that are designed better for capital letters. These alternates are flatter. GSUB lookups 23 and 24 implement the substitution. This substitution is not fired on some capital consonants (T, V, W, X) because these marks are not expected on these bases.

Libertinus has custom glyphs for some marks that are designed better for superscripts. These alternates are smaller. Regrettably?, no GSUB table implements their substitution.

A circumflex Â Ã Ä Å Æ

O circumflex Ô Õ Ö Ø Ñ

AE ligature Æ Æ Æ Æ Æ

Š Ţ Ŧ Š Ŧ Š Ŧ Š Ŧ Ŧ

Š Ŧ Ŧ Š Ŧ Š Ŧ Š Ŧ Ŧ

Regular, above-anchor, capitals, *X* coordinate

Figure 3 (left panel) shows pairs of bases aligned at their above-anchor. If a capital letter has a bad *X* coordinate for its above-anchor, then it will look always off, right or left, in its black column relative to the gray letters that it overlays. Note that the above-anchor *X* coordinate is not in the geometric (horizontal) center of the glyph, by design. The above-anchor *X* coordinate is always off center by a few UPM units, sometimes left and sometimes right. But it very hard to discern a few UPM units. See the right panel of Figure 3 for an illustration of increasingly bad (spoofed) *X* coordinates of N's above-anchor.

Note that the above-anchor of L is above the major (vertical) stem of the L. For E, F, P and R, the above-anchor is optically centered above the glyph, not above the major stem.

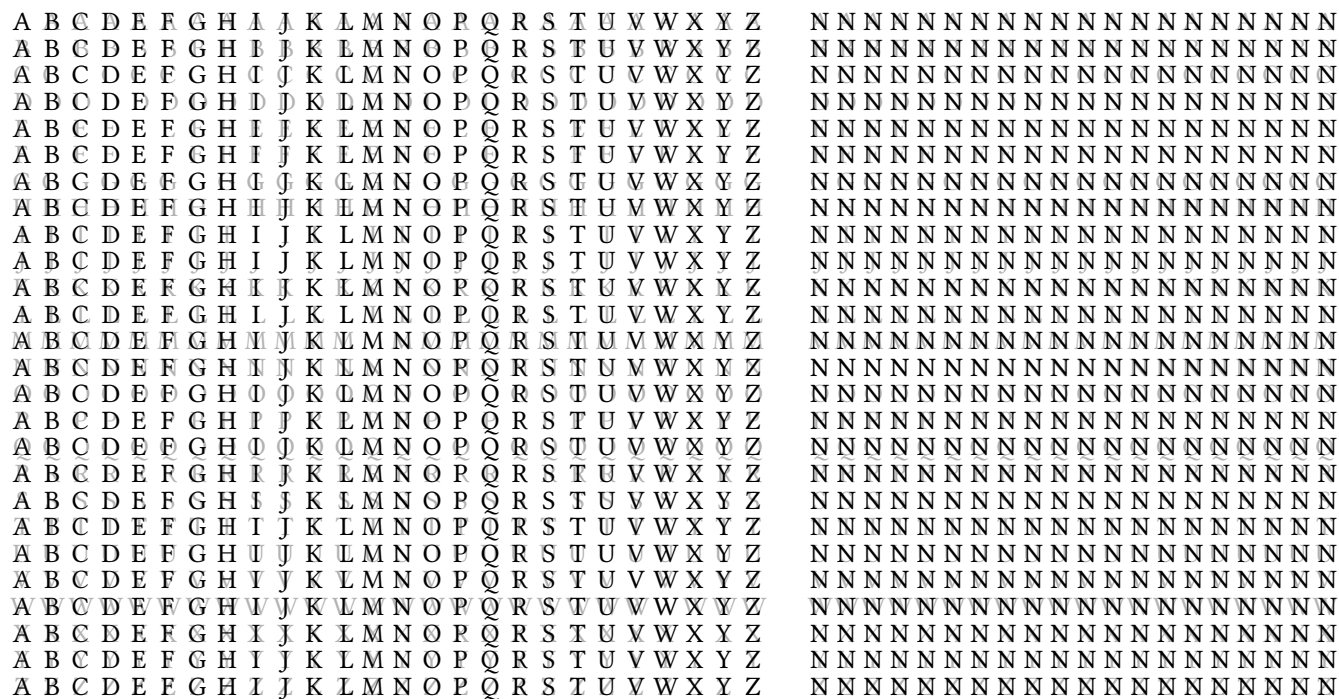


Figure 3: Above-anchor X coordinate alignment. In the left panel, the 26×26 grid shows letters A–Z aligned at their above-anchor. The right panel shows how hard it is to discern bad X coordinates. The right panel shows N aligned with letters at their above-anchor, starting at the first column with Ns above-anchor as currently set by the font, and then spoofing increasing bad values for the X coordinate of N’s above-anchor, by 5 UPM per column. At the rightmost column, it becomes evident that the (spoofed) X coordinate of N’s above-anchor is bad, because in that column the black Ns are too far right of the gray letters below them.

Comma-style marks

Libertinus has some (precomposed) glyphs for Unicode characters (with code points) where the caron or cedilla has a comma-style shape, instead of a wedge or hook as might be naively expected. But all of these are appropriate for Czech/Slovak orthography (caron or *háček*) or Latvian orthography (cedilla, but really the comma-above mark or *komatiņš*, by a known mistake in Unicode).

Czech/Slovak *háček*

In Czech/Slovak orthography, the d, l, L, n and t may take a reduced caron that may look like a comma, not a wedge. In Libertinus, only the d, L, l, and t have the reduced comma-style caron in their precomposed form, and the n has wedge caron precomposed form.

`\char"010F` **ď** *ď* **ď** *ď*

`\char"013D` **Ľ** *Ľ* **Ľ** *Ľ*

`\char"013E` **ĺ** *ĺ* **ĺ** *ĺ*

`\char"0148` **ň** *ň* **ň** *ň*

`\char"0165` **ť** *ť* **ť** *ť*

Libertinus even has a custom glyph for a small capital L with a caron, and it is comma-style.

`\textsc{1\char"030C}` **ɾ**

`\textsc{\char"013E}` **ɽ**

Cedilla and Latvian *komatiņš*

Here are all precomposed characters with a cedilla.

`\char"00C7` **Ç** *Ç* **Ç** *Ç* `\char"00E7` **ç** *ç* **ç** *ç*

`\char"1E08` **Ĉ** *Ĉ* **Ĉ** *Ĉ* `\char"1E08` **ĉ** *ĉ* **ĉ** *ĉ*

`\char"1E10` **Ď** *Ď* **Ď** *Ď* `\char"1E11` **ď** *ď* **ď** *ď*

`\char"0228` **Ē** *Ē* **Ē** *Ē* `\char"0229` **ē** *ē* **ē** *ē*

`\char"1E1C` **Ě** *Ě* **Ě** *Ě* `\char"1E1D` **ě** *ě* **ě** *ě*

`\char"0122` **Ĝ** *Ĝ* **Ĝ** *Ĝ* `\char"0123` **ĝ** *ĝ* **ĝ** *ĝ*

`\char"1E28` **Ĥ** *Ĥ* **Ĥ** *Ĥ* `\char"1E29` **ĥ** *ĥ* **ĥ** *ĥ*

`\char"0136` **Ķ** *Ķ* **Ķ** *Ķ* `\char"0137` **ķ** *ķ* **ķ** *ķ*

`\char"013B` **Ļ** *Ļ* **Ļ** *Ļ* `\char"013C` **ļ** *ļ* **ļ** *ļ*

`\char"0145` **Ņ** *Ņ* **Ņ** *Ņ* `\char"0146` **ņ** *ņ* **ņ** *ņ*

`\char"0156` **Ŗ** *Ŗ* **Ŗ** *Ŗ* `\char"0157` **ŗ** *ŗ* **ŗ** *ŗ*

`\char"015E` **Ș** *Ș* **Ș** *Ș* `\char"015F` **ș** *ș* **ș** *ș*

`\char"0162` **Ț** *Ț* **Ț** *Ț* `\char"0163` **ț** *ț* **ț** *ț*

These characters – Ģ ģ Ķ ķ Ļ ļ Ņ ņ Ŗ ŗ – use the Latvian *komatiņš* which is proper for Latvian orthography, but the result of a historical mistake in Unicode naming.

The Unicode characters of D and d with cedilla exist for scholarly and historical reasons, not Latvian orthography. Fonts design these Unicode characters differently. Libertinus designs them inconsistently: comma-style in regular and italic, and hook-style in (semi)bold and (semi)bold italic. I think the cedilla on a D or d should be hook-like, but I do not really know.

The other characters – c, e, h, s, t – with a cedilla should definitely be hook-like, and they are so in Libertinus.

Somewhat mysteriously, if the LaTeX source tries to combine any of these letters – c, d, e, g, h, k, l, n, r, s, t – with the combining cedilla, U+0327, then the precomposed character is produced.

This behavior does not make sense for the Latvian *komatiņš* bases – g, k, l, n, r – because any person trying to produce Latvian g with *komatiņš* would not try $g\backslash\text{char}^{\text{"0327"}}$ or $\backslash c\{g\}$. This behavior is tolerable for the other bases – c, d, e, h, s, t.

The ij and IJ ligatures are Unicode characters (with code points) and are supported by Libertinus.

ij \char"0133 U+0133 small ligature ij

`\newfontfamily\SerifRegularDutch[Language = Dutch]{LibertinusSerif-Regular.otf}`

íj {\SerifRegularDutch \char"0133\char"0301} Dutch small ligature ij with acute

ij {\SerifRegularDutch \textsc{\char"0133\char"0301}} Dutch small capital ligature IJ with acute

In an English LaTeX document, `ij` and `IJ` are kerned by the GPOS table to produce `ij` and `IJ`, so they look close together, but they are not substituted by the single glyph for the ligature.

Consider the tilde below. There is the combining tilde below, U+0330, encoded in LaTeX as the codepoint `\char"0330`. Remember to put the base before the combining mark in LaTeX, e.g. `a\char"0330` a. Three vowels (e, i, u) have precomposed characters – lowercase and capital – encoded in LaTeX as these codepoints:

E, e, I, i, U, or u followed by `\char"0330` will produce the precomposed character glyph. A, a, O, and o, as well as any other IPA vowel with a below-anchor, followed by `\char"0330` will be shaped by attaching the combining below tilde to that vowel at their below-anchor. Note the specific case of i. `\char"0330 = \char"1E2D i = i` is a dotted i (a precomposed glyph).

Suppose you want a vowel with a tilde below, possibly also with a mark above, especially acute, grave, caron, double acute, or double grave. In the specific case of i, the above mark will crash into the dotted i unless you take care to encode the triplet starting with a dotless i, U+0131. For example, consider the combining acute U+0301.

```
\char"0131\char"0330\char"0301 í í í í
```

Recall that the canonical order for a triplet of a base and two combining marks (one below and one above) is base + combining below mark + combining above mark, not base + above + below.

Consider the wrong order:

```
i\char"0301\char"0330 i i i i\char"0301 i i i i\char"0330 i i i i
\char"00EC\char"0330 i i i i\char"00EC i i i i
\char"1E2D\char"0301 i i i i\char"1E2D i i i i
```

But note that the position of the combining tilde below is bad in italic and semibold italic. This is because the below-anchor for the combining tilde below mark is bad. For tilde below, the precomposed character glyphs with i, e, and u are sometimes good and sometimes bad in italic and semibold italic.

Precomposed characters are blue and therefore cannot be changed unless their glyph outline is changed. Black combinations of vowels and marks are shaped by anchors and therefore can be changed by setting good anchors.

Use the combining grapheme joiner (CGJ) character U+034F, \char"034F, or \cgj if loaded by some package, to force the shaper to use combining marks instead of precomposed glyphs.

```
AEIOUE\char"034Faeiouε\char"034F AEIOUE\char"034Faeiouε\char"034F AEIOUE\char"034Faeiouε\char"034F AEIOUE\char"034Faeiouε\char"034F
ÁÉÍÓÚÊ\char"034Fáéíóúés\char"034F ÁÉÍÓÚÊ\char"034Fáéíóúés\char"034F ÁÉÍÓÚÊ\char"034Fáéíóúés\char"034F ÁÉÍÓÚÊ\char"034Fáéíóúés\char"034F
ÀÈÌÒÙË\char"034Fàèìòùè\char"034F ÀÈÌÒÙË\char"034Fàèìòùè\char"034F ÀÈÌÒÙË\char"034Fàèìòùè\char"034F ÀÈÌÒÙË\char"034Fàèìòùè\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
```

Patched

```
AEIOUE\char"034Faeiouε\char"034F AEIOUE\char"034Faeiouε\char"034F AEIOUE\char"034Faeiouε\char"034F AEIOUE\char"034Faeiouε\char"034F
ÁÉÍÓÚÊ\char"034Fáéíóúés\char"034F ÁÉÍÓÚÊ\char"034Fáéíóúés\char"034F ÁÉÍÓÚÊ\char"034Fáéíóúés\char"034F ÁÉÍÓÚÊ\char"034Fáéíóúés\char"034F
ÀÈÌÒÙË\char"034Fàèìòùè\char"034F ÀÈÌÒÙË\char"034Fàèìòùè\char"034F ÀÈÌÒÙË\char"034Fàèìòùè\char"034F ÀÈÌÒÙË\char"034Fàèìòùè\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F ĂĖĬŎŨ\char"034Făėĭŏũ\char"034F
```

ĀĒĪŌŪẼǾǻēīōūěǿ ĀĒĪŌŪẼǾǻēīōūěǿ ĀĒĪŌŪẼǾǻēīōūěǿ ĀĒĪŌŪẼǾǻēīōūěǿ
 ĀĒĪŌŪẼǾǻēīōūěǿ ĀĒĪŌŪẼǾǻēīōūěǿ ĀĒĪŌŪẼǾǻēīōūěǿ ĀĒĪŌŪẼǾǻēīōūěǿ
 ĀĒĪŌŪẼǾǻēīōūěǿ ĀĒĪŌŪẼǾǻēīōūěǿ ĀĒĪŌŪẼǾǻēīōūěǿ ĀĒĪŌŪẼǾǻēīōūěǿ
 ĀĒĪŌŪẼǾǻēīōūěǿ ĀĒĪŌŪẼǾǻēīōūěǿ ĀĒĪŌŪẼǾǻēīōūěǿ ĀĒĪŌŪẼǾǻēīōūěǿ
 ĀĒĪŌŪẼǾǻēīōūěǿ ĀĒĪŌŪẼǾǻēīōūěǿ ĀĒĪŌŪẼǾǻēīōūěǿ ĀĒĪŌŪẼǾǻēīōūěǿ
 ĀĒĪŌŪẼǾǻēīōūěǿ ĀĒĪŌŪẼǾǻēīōūěǿ ĀĒĪŌŪẼǾǻēīōūěǿ ĀĒĪŌŪẼǾǻēīōūěǿ

Mark over space

I added a new en-width x-height PUA character, U+E100, with anchors. To float an above mark, say double grave U+030F over a space, write `\char"E100\char"030F` But in practice, the floating mark carries the same tonal specification as the preceding vowel, and therefore must align exactly with the height of that vowel's combining mark. To ensure this mark alignment, insert a combining grapheme joiner, `\cgj`, in the prior vowel + mark combo, to force anchor positioning without substituting a precomposed glyph there. Note that in Libertinus, the two graves in the U+030F glyph are not translates of an identical outline; this is a deliberate design choice by the designer.

Original

`-ke\char"030F\ \char"030F -kë" -kë" -kë" -kë"`

Patched without CGJ

`-ke\char"030F \char"E100\char"030F -kë" -kë" -kë" -kë"`

Patched with CGJ

`-ke\cgj\char"030F \char"E100\char"030F -kë" -kë" -kë" -kë"`

f with dot below

tiyó ṛfa

tiyó ṛfa

tiyó ṛfa

tiyó ṛfa

`ṛ` = `\char"1E5B` is the precomposed character U+1E5B Latin small letter r with dot below.

`f̣` = `f\char"0323` is f with U+0323 combining dot below.

Here are some letters with a dot below.

a b d e h i k l m n o r s t u v w y z c f g j p q x

a b d e h i k l m n o r s t u v w y z c f g j p q x

a b d e h i k l m n o r s t u v w y z c f g j p q x

a b d e h i k l m n o r s t u v w y z c f g j p q x

The left group of letters – a b d e h i k l m n o r s t u v w y z – have a precomposed character. For the right group of letters – c f g j p q x – the dot is placed according to the below-anchor (if the below-anchor is defined for the letter) or by HarfBuzz fallback shaping (placing the dot below the baseline, regardless of the letter's bounding box).

In all styles – regular, italic, semibold, and semibold italic – the precomposed glyph of y with dot below places the dot beside the descender, even though the y has a below-anchor below the descender.

Do the letters c, f, g, j, p, q, and x have good below-anchors? In regular, the below-anchor of all these combining bases is below the glyph outline (good). In italic, the below-anchor of f, p, and q is beside the descender (bad), the below-anchor of g and j is below the descender (good), and the below-anchor of c and x is below the baseline (good). In semibold, the below-anchor of g and j is below the descender (good), but the other bases – c, f, p, q – have no below-anchor, and HarfBuzz fallback positioning places the combining dot below the baseline, not below the bounding box; for c, f and x this is good, but for p and q this is bad, because of the descenders. In semibold italic, the below-anchor of c, g, j, p and q is below the glyph outline (good), the below-anchor of f is inside the descender (very bad), and x has no below-anchor but HarfBuzz fallback positioning places the combining dot below the baseline (good).

Which dots are used in precomposed characters and how big are the dots? In all styles (except semibold italic), the precomposed characters reference the dot of uni0307, combining dot above, U+0307, and take care to translate this above-mark to a position below the letter outline, except for m, which references dotbelowcomb, combining dot below, U+0323. In semibold italic, dotaccent is used instead of uni0307. These two dots are four-point Bézier circles (in regular and semibold, but ovals in italic and semibold italic), but of different diameter. In regular, uni0307 has diameter 100, and dotbelowcomb has diameter 108. That is, the precomposed characters (except m) use a smaller dot, and the non-precomposed letters (and m) take a bigger dot. In semibold, the size difference is reversed: uni0307 has larger diameter 124, and dotbelowcomb has small diameter 108. Actually, semibold dotbelowcomb is not a Bézier circle – the horizontal span is 109, and the vertical span is 108. In semibold italic, dotaccent is a skewed four-point Bézier oval – the horizontal and vertical spans are 120, offset by $(\pm 6, \pm 7.5)$ from the center – and dotbelowcomb is slightly larger, slightly flatter skewed six-point Bézier oval – horizontal span is 122 and the vertical span is 120. So, in semibold italic the two dots in precomposed and combining form are closest in size. Recall that f has a bad below-anchor, inside the descender, but g, j, p and q have a good below-anchor, below the glyph outline.

So, in order to solve the problem of the bad dot below f in italic, I need to decide the scope of the badness to correct. The simplest patch is move the below-anchor in italic and semibold italic below the descender. But this does not solve inconsistent below dot size across styles and letters. In italic and semibold italic, g and j have a (good) below-anchor at $Y = -319$, so that is also good Y coordinate for f's below-anchor.

I asked Copilot/GPT-5.1 to calculate a good X coordinate for f's below-anchor in italic and semibold italic, taking into account the U-pocket of f's descender and the below-anchor of dotbelowcomb. I decided that this U-pocket strategy was appropriate after testing other strategies that accounted for the slant of onf the major stroke. Here is a reasonably faithful summary of the calculations and reasoning by Copilot/GPT-5.1.

For italic, the descender's U-pocket is defined by the control points at (14, -238) and (33, -205), giving a U-center around $x \approx 24$. Using the (symmetric) italic dotbelowcomb anchor as the dot's optical center, the dot should sit under this U-pocket, not under the ball or the stem. A balanced, U-centered, slightly inner-biased choice is $X \approx 30$.

For semibold italic, the U-pocket of the semibold italic descender is wider and more asymmetric, with a geometric U-center of $X = -33.5$. The semibold italic dotbelowcomb itself is also skewed, with its anchor already shifted relative to its mass. In practice, the visually correct position for the dot below f is under the open side of the U-pocket, clearly right of the ball and closer the right of the stem. The anchor candidate is $X \approx 62$.

Here are examples with the patched fonts, with f's below-anchor in italic at (30, -319) and semibold italic at (62, -319). This does not solve the problem of inconsistent dot size below letters, and I did not touch p, q, or y.

Original Patched italics (f only)

tiyó rfa	tiyó rfa
tiyó rfa	tiyó rfa
tiyó rfa	tiyó rfa
tiyó rfa	tiyó rfa

Original (repeated from above)

a b d e h i k l m n o r s t u v w y z	c f g j p q x
a b d e h i k l m n o r s t u v w y z	c f g j p q x
a b d e h i k l m n o r s t u v w y z	c f g j p q x
a b d e h i k l m n o r s t u v w y z	c f g j p q x

Patched italics (f only)

a b d e h i k l m n o r s t u v w y z	c f g j p q x
---------------------------------------	---------------

a b d e h i k l m n o r s t u v w x z c f g j p q x

Patch superscript consonant anchors

Libertinus serif regular

h ħ j f r x ı ʒ ʃ w y ʋ ł s x ʁ b k m p t u v œ β γ δ

h ħ j f r x ı ʒ ʃ w y ʋ ł s x ʁ b k m p t u v œ β γ δ

h ħ j f r x ı ʒ ʃ w y ʋ ł s x ʁ b k m p t u v œ β γ δ

h ħ j f r x ı ʒ ʃ w y ʋ ł s x ʁ b k m p t u v œ β γ δ

h ħ j f r x ı ʒ ʃ w y ʋ ł s x ʁ b k m p t u v œ β γ δ

h ħ j f r x ı ʒ ʃ w y ʋ ł s x ʁ b k m p t u v œ β γ δ

Libertinus serif regular, patched

h ħ j f r x ı ʒ ʃ w y ʋ ł s x ʁ b k m p t u v œ β γ δ

h ħ j f r x ı ʒ ʃ w y ʋ ł s x ʁ b k m p t u v œ β γ δ

h ħ j f r x ı ʒ ʃ w y ʋ ł s x ʁ b k m p t u v œ β γ δ

h ħ j f r x ı ʒ ʃ w y ʋ ł s x ʁ b k m p t u v œ β γ δ

h ħ j f r x ı ʒ ʃ w y ʋ ł s x ʁ b k m p t u v œ β γ δ

h ħ j f r x ı ʒ ʃ w y ʋ ł s x ʁ b k m p t u v œ β γ δ

Gentium Plus

h ħ j f r x ı ʒ ʃ w y ʋ ł s x ʁ b k m p t u v œ β γ δ

h ħ j f r x ı ʒ ʃ w y ʋ ł s x ʁ b k m p t u v œ β γ δ

h ħ j f r x ı ʒ ʃ w y ʋ ł s x ʁ b k m p t u v œ β γ δ

h ħ j f r x ı ʒ ʃ w y ʋ ł s x ʁ b k m p t u v œ β γ δ

h ħ j f r x ı ʒ ʃ w y ʋ ł s x ʁ b k m p t u v œ β γ δ

h ħ j f r x ı ʒ ʃ w y ʋ ł s x ʁ b k m p t u v œ β γ δ

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z a b c d e f g h i j k l m n o p q r s t u v w x y z Æ æ Œ œ Ø ø Đ đ Þ þ ß Ŧ ŧ Đ đ Ħ ħ ı ĵ Ł ł Ɓ Ƀ

À Á Â Ã Ä Å Æ Ç È É Ê Ë Ì Í Î Ï Ñ Ò Ó Ô Õ Ö × Ø Ù Ú Û Ü Ý Þ ß à á â ã ä å æ ç è é ê ë ì í î ï ñ ò ó ô õ ö ÷ ø ù ú û ü ý þ ß Ŧ ŧ Đ đ Ħ ħ ı ĵ Ł ł Ɓ Ƀ

Dot removal

To do. Check GSUB to see if dot removal for i and j is complete.

Precomposed characters

To do. Check GSUB if an available precomposed character is encoded by a doublet of the base and a combining mark, or a triplet of the base and the two combining marks, in either mark order, then the precomposed character should be produced by the font's GSUB table.

Marks

Here is a catalog of combining marks in Unicode, grouped by the seven GPOS anchor class indices in Libertinus (0–6).

Above marks – GPOS anchor 0

x
x
x
x x

Below marks – GPOS anchor 2

X
X
X
X X

Left angle – GPOS anchor 3

$$\mathbf{X}^T \mathbf{X}^T \bar{\mathbf{X}} \bar{\mathbf{x}}$$

Below-right marks – GPOS anchor 4

$$\begin{array}{c} \mathbf{x}_j \mathbf{x}_l \mathbf{x}_z \\ \mathbf{x}_j \mathbf{x}_l \mathbf{x}_z \\ \mathbf{x}_j \mathbf{x}_l \mathbf{x}_z \\ \mathbf{x}_j \mathbf{x}_l \mathbf{x}_z \end{array}$$

Cedilla – GPOS anchor 5

$$\mathbf{X}_3 \quad \mathbf{x}_3 \quad \mathbf{X}_3 \quad \mathbf{x}_3$$

Overlay marks – GPOS anchor 6

$\emptyset \emptyset \emptyset \emptyset \emptyset$
 $\emptyset \emptyset \emptyset \emptyset \emptyset$
 $\emptyset \emptyset \emptyset \emptyset \emptyset$
 $\emptyset \emptyset \emptyset \emptyset \emptyset$

Double marks

xx xx xx xx xx xx xx
xx xx xx xx xx xx xx
xx xx xx xx xx xx xx
xx xx xx xx xx xx xx
xx xx xx xx xx xx xx

Greek marks

τ τ τ τ τ τ τ
τ τ τ τ
τ τ τ τ τ τ τ
τ τ τ τ τ τ τ
τ τ τ τ τ τ τ
τ τ τ τ τ τ τ
τ τ τ τ τ τ τ
τ τ τ τ τ τ τ
τ τ τ τ τ τ τ
τ τ τ τ τ τ τ
τ τ τ τ τ τ τ

Overline and low line

U+0305 combining overline and U+0332 combining low line are applied by GSUB lookups that substitute a custom-width line mark according to width classes. GSUB lookup 51 effectively classifies the glyphs by width, and GSUB lookups 52–82 provide the appropriate custom-width mark for the base width class. Semibold and semibold italic do not have any combining low lines. Custom-width line marks do not have anchors. They are positioned by HarfBuzz fallback shaping. The overlines are above the ascender line, so they are above all bases. The low lines are below the baseline, but above the descender line, and so they cut through any descenders. If a new glyph is added, and you want to support custom-width line marks for it, then remember to classify the new glyph in GSUB lookup 51.

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæØøÐðÞþǷǷǺǺǻǻǼǼǾǾǿǿǠǠǡǡǢǢǣǣǤǤǥǥǦǦǧǧǨǨǩǩǪǪǫǫǬǬǭǭǮǮǯǯǰǰǱǱǲǲǳǳǴǴǵǵǶǶǷǷǸǸǹǹǺǺǻǻǼǼǾǾǿǿǠǠǡǡǢǢǣǣǤǤǥǥǦǦǧǧǨǨǩǩǪǪǫǫǬǬǭǭ

Above-right marks – GPOS anchor 1

Libertinus has three combining marks with GPOS anchor 1 – U+031B horn, U+0315 comma above right, and U+0358 dot above right.

A few bases in Libertinus have this anchor (anchor 1), but none that possibly need it.

Horn

The Vietnamese alphabet has two horn vowels – Ō ơ Ūư – and these vowels may take tonal marks. All these combinations are Unicode characters (with code points).

ŌōƯưỒồỖỗỖỗỒồƯứỪừỦủỮữƯư

O o U u Ö ö Õ õ Ö ö Õ õ O o Ú ú Û ù Ũ ñ Ũ ñ U u

Ó ó Ů ů Ő ő Œ œ Œ œ Ů ů Ű ű Ų ų Ŵ ŵ

O o U u Ó ó Ò ò Ô ô Õ õ Ö ö Ø ø Ù ú Û û Ü ü U u

There is a Unicode character for the combining horn mark: `\char"031B U+031B`, but this is not used in practice. The Vietnamese vowels are usually entered in a LaTeX document as Unicode characters, also with `\usepackage[vietnamese]{babel}` for hyphenation and spacing.

Comma above right

There is a Unicode character for the combining comma above right mark: `'\char"0315 U+0315`.

U+0315 (comma above right) is used in rare cases when another comma-like mark does not fit.

In IPA, the glottalization diacritic mark uses U+0313 (comma above) as a combining mark. But when a glottalization mark is needed for a superscript consonant (1) or superscript consonant cluster (2), then the combining mark U+0315 (comma above right) is used instead, because its positioning is hopefully better than that of U+0313.

(1) *kʷkt* [*kʷkʰt*] ‘to scrape’ (Nuxalk)

(2) *p'ak* [p^{kx}ak] 'he/she cut it' (Upper Chehalis)

In (1), the superscript consonant cluster is ^{wkx} and it is glottalized. In (2), the superscript consonant ^x is glottalized.

Because U+0315 is being used in superscript position on a smaller sized glyph, U+0315 should probably be smaller than U+0313, but it is not so in Libertinus.

Dot above right

There is a Unicode character for the combining dot above right mark: `\char"0358` U+0358 dot above right

U+0358 should be a small, above-right dot, used to mark palatalization or similar secondary articulations on superscript consonants, created for the same typographic reason U+0315 exists – the standard palatalization diacritic (U+0307, dot above) is too big and too high.

In Libertinus, U+0358 is a 104-diameter Bezier circle, and U+0307 is a smaller Bezier oval, inverting the desired size difference.

There are only a few IPA superscript letters that may serve as the possible host of U+0315 or U+0358 on superscript consonants or on superscript consonant clusters. In Libertinus, regrettably none of these characters has the anchor (anchor 1), so the mark (U+0315 and U+0358) is positioned by HarfBuzz fallback shaping, which seems to place the mark at the right boundary of the glyph at a constant vertical aspect (effectively at the superscript x-height).

These IPA marks U+0315 and U+0358 should only be used in regular (not italic and not semibold). Regrettably, in Libertinus serif semibold, U+0358 looks like a grave, not a dot, and there is no U+0358 glyph in Libertinus serif semibold italic.

There are three comma/apostrophe glyphs that could possibly mark glottalization: U+0313 combining comma above, U+0315 combining comma above right, and U+02BC modifier letter apostrophe. In Libertinus, U+0315 has the worst appearance, even though U+0315 is intended for superscript letters. In Gentium Plus (SIL Global), U+0315 is tightly above-right the superscript letter, as expected.

Libertinus

[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t] [k^{wkx̣}t]

[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t] [k^{wkx̣}t]

[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t] [k^{wkx̣}t]

Gentium Plus

[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t] [k^{wkx̣}t]

[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t] [k^{wkx̣}t]

[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t] [k^{wkx̣}t]

I patched Libertinus to add anchors to those IPA superscript letters to support U+0315.

Patched Libertinus, selected IPA superscript letters with U+0315 and U+0358

ḥḥj̣ṛɹ̣ɹ̣ɹ̣ɹ̣ẉỵỵḷṣx̣ḅḳṃp̣ṭụṿ.ɹ̣β̣ɣ̣δ̣

ḥḥj̣ṛɹ̣ɹ̣ɹ̣ẉỵỵḷṣx̣ḅḳṃp̣ṭụṿ.ɹ̣β̣ɣ̣δ̣

[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t] [k^{wkx̣}t]

Gentium Plus, selected IPA superscript letters with U+0315 and U+0358

ḥḥj̣ṛɹ̣ɹ̣ɹ̣ẉỵỵḷṣx̣ḅḳṃp̣ṭụṿ.ɹ̣β̣ɣ̣δ̣

ḥḥj̣ṛɹ̣ɹ̣ɹ̣ẉỵỵḷṣx̣ḅḳṃp̣ṭụṿ.ɹ̣β̣ɣ̣δ̣

[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t] [k^{wkx̣}t]

Here are the IPA superscript letters considered and their new anchor (anchor 1).

U+02B0: (347, 648), U+02B1: (339, 648), U+02B2: (193, 648), U+02B3: (284, 648), U+02B4: (274, 648), U+02B5: (319, 648), U+02B6: (308, 648), U+02B7: (497, 648), U+02B8: (368, 648), U+02E0: (341, 648), U+02E1: (193, 648), U+02E2: (270, 648), U+02E3: (344, 648), U+02E4: (282, 648), U+1D47: (319, 648), U+1D4F: (353, 648), U+1D50: (533, 648), U+1D56: (334, 648), U+1D57: (241, 648), U+1D5A: (525, 648), U+1D5B: (354, 648), U+1D5C: (508, 648), U+1D5D: (335, 648), U+1D5E: (372, 648), U+1D5F: (353, 648)

I did not change the anchor-1 of U+315 and U+0358. U+0358 has its anchor at the xMin and yMin of its bounding box. But U+0315 already has some clearance built in to its anchor: its anchor is 62 units below its yMin, but 25 units right of it xMin. So, I decided to provide x-clearance of 12 units and y-clearance of 80 units from the bases. All these IPA superscript letters already have a superscript meanline at $Y = 630$. So their anchor will be at $Y = 630 - 62 + 80 = 648$. The X coordinate depends on their bounding box and its xMax. To provide the desired clearance of 12 units, the anchor is at $X = xMax + 26 + 12$. These anchor coordinate values were optimized for U+0315 but they work fine for U+0358.

Bases

Here are some subgroups of bases.

Latin

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæƎœØøÐðÞþßŒœĐđĤĥıŁłǼ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæƎœØøÐðÞþßŒœĐđĤĥıŁłǼ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæƎœØøÐðÞþßŒœĐđĤĥıŁłǼ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæƎœØøÐðÞþßŒœĐđĤĥıŁłǼ

IPA

b0C0D0D0q0E0F0G0Yh1KWN03211#1e0a0b0c0d0e0s0e33e0f0g0Y0y0h0i0u0t0l0k0w0y0m0j0n0e0w0f0J0I0r0r0R0K0S0s0f0l0t0u0S0U0M0A0Y0Z0z0Z030Y0J0C0B0e0H0j0X0L0s
 b0C0D0D0q0E0F0G0Yh1KWN03211#1e0a0b0c0d0e0s0e33e0f0g0Y0y0h0i0u0t0l0k0w0y0m0j0n0e0w0f0J0I0r0r0R0K0S0s0f0l0t0u0S0U0M0A0Y0Z0z0Z030Y0J0C0B0e0H0j0X0L0
 b0C0D0D0q0E0F0G0Yh1KWN03211#1e0a0b0c0d0e0s0e33e0f0g0Y0y0h0i0u0t0l0k0w0y0m0j0n0e0w0f0J0I0r0r0R0K0S0s0f0l0t0u0S0U0M0A0Y0Z0z0Z030Y0J0C0B0e0H0j0X0L0
 b0C0D0D0q0E0F0G0Yh1KWN03211#1e0a0b0c0d0e0s0e33e0f0g0Y0y0h0i0u0t0l0k0w0y0m0j0n0e0w0f0J0I0r0r0R0K0S0s0f0l0t0u0S0U0M0A0Y0Z0z0Z030Y0J0C0B0e0H0j0X0L0

Superscript consonants

hñj r l d b w y y l s x f b k m p t u v ѿ β γ δ
hñj r l d b w y y l s x f b k m p t u v ☒☒☒☒
hñ j r l d b w y y l s x f b k m p t u v ☒☒☒☒
hñj r l d b w y y l s x f b k m p t u v ☒☒☒☒

Greek

$\text{AB}\Gamma\Delta\text{E}\text{Z}\text{H}\Theta\text{I}\text{K}\text{L}\text{M}\text{N}\Xi\text{O}\Pi\rho\sigma\tau\upsilon\phi\chi\psi\Omega\alpha\beta\gamma\delta\epsilon\zeta\eta\theta\iota\kappa\mu\nu\xi\omicron\pi\rho\sigma\tau\upsilon\phi\chi\psi\omega$
 $\text{AB}\Gamma\Delta\text{E}\text{Z}\text{H}\Theta\text{I}\text{K}\text{L}\text{M}\text{N}\Xi\text{O}\Pi\rho\sigma\tau\upsilon\phi\chi\psi\Omega\alpha\beta\gamma\delta\epsilon\zeta\eta\theta\iota\kappa\mu\nu\xi\omicron\pi\rho\sigma\tau\upsilon\phi\chi\psi\omega$
 $\text{AB}\Gamma\Delta\text{E}\text{Z}\text{H}\Theta\text{I}\text{K}\text{L}\text{M}\text{N}\Xi\text{O}\Pi\rho\sigma\tau\upsilon\phi\chi\psi\Omega\alpha\beta\gamma\delta\epsilon\zeta\eta\theta\iota\kappa\mu\nu\xi\omicron\pi\rho\sigma\tau\upsilon\phi\chi\psi\omega$
 $\text{AB}\Gamma\Delta\text{E}\text{Z}\text{H}\Theta\text{I}\text{K}\text{L}\text{M}\text{N}\Xi\text{O}\Pi\rho\sigma\tau\upsilon\phi\chi\psi\Omega\alpha\beta\gamma\delta\epsilon\zeta\eta\theta\iota\kappa\mu\nu\xi\omicron\pi\rho\sigma\tau\upsilon\phi\chi\psi\omega$

Cyrillic

АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїŸÿĂăĖėĮį

АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯ*абвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїŸÿĂăĖėĮį*

АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯ**абвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїŸÿĂăĖėĮį**

АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯ*абвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїŸÿĂăĖėĮį*

[illegible][illegible][illegible][illegible]

No marks: rare, digraph etc.

Σ ξ D Z D z d z D Ž D ž d ž L J L j l j N J N j n j

 $\Xi_{\mathcal{E}} D Z D z d z D \check{Z} D \check{z} d \check{z} L J L j l j N J N j n j$

Σ ξ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj

Ξ ξ D Z D z d z D Ž D ž d ž L Ĵ L j l j N J N j n j

Small capitals

ABCDEFGHIJKLMNOPQRSTUVWXYZÆÐŒÞIJßŋ

ABCDEFGHIJKLMNOPQRSTUVWXYZÆÐƆP IJβη

ABCDEFGHIJKLMNOPQRSTUVWXYZÆÐËÞIJßŋ

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Æ Þ I J ß ŋ

Precomposed characters of anchor-relevant bases and marks

àáâãäåääääääääääääèéêëēẽēēěëïîíĩìòóôõōöőűúûũüýÿŷÿȳđłŋřšţ

à á â ã ä å æ ç è é ê ë ì í î ï ð ó ô õ ö ø ù ú û ü ý ÿ ÿ d l n r s t z

àáâãäåæçèéêëìíîïðñòóôõö÷øùúûüýÿđłŋŗţȚ

à á â ã ä å æ ç è é ê ë ì í î ï ð ó ô õ ö ø ö ő û ü ū ū ŷ ý ÿ ÿ y d l n r s t z

Base and mark shaping – Linguistics notation

These grids display base and mark rendering in XeLaTeX documents in Libertinus serif regular, italic, semibold and semibold italic.

The grids use Latin in IPA bases and combining marks that are used in linguistics notation.

A black color means GPOS anchor positioning.

A blue color means a precomposed glyph.

A red color means HarfBuzz fallback shaping (no anchors, no precomposed glyph, no GSUB substitution).

A faded color – gray, light blue, or pink – means that the base and mark combination is not reasonably expected in notation in linguistics.

A gray color can also mean that mark glyph is missing in the font. In this case, just the gray base is printed. This only happens in Libertinus Serif Semibold and Semibold Italic for three below marks: ̐ (U+0333 double macron below), ̑ (U+033A inverted bridge below), and ̒ (U+033B square below). Otherwise, all considered base and mark combinations can be drawn in XeLaTeX documents in Libertinus without tofu for missing glyphs.

Regular, Bases requiring anchors for common marks, Common combining marks

[illegible]

Regular patched, Bases requiring anchors for common marks, Common combining marks

[illegible]

[illegible]

Italic patched, Bases requiring anchors for common marks, Common combining marks

[illegible]

Semibold, Bases requiring anchors for common marks, Common combining marks

[illegible]

Semibold patched, Bases requiring anchors for common marks, Common combining marks

[illegible]

[illegible]

Semibold italic patched, Bases requiring anchors for common marks, Common combining marks

à B c d e f g i j k l m n ò p q r s t u v w x y z æ œ ñ è à ò b d d é é é j g g è y y q h l l l k l u m j n ñ ò è i j l r r s s f u à à y z z j l
á B c d e f g i j k l m n ó p q r s t u v w x y z æ œ ñ é á ó b d d é é é j g g è y y q h l l l k l u m j n ñ ó è i j l r r s s f u á á y z z j l
â B c d e f g i j k l m n ô p q r s t u v w x y z æ œ ñ ê â ô b d d é é é j g g è y y q h l l l k l u m j n ñ ô è i j l r r s s f u â â y z z j l
ã B c d e f g i j k l m n õ p q r s t u v w x y z æ œ ñ ã õ ã b d d é é é j g g è y y q h l l l k l u m j n ñ õ è i j l r r s s f u ã ã y z z j l
ä B c d e f g i j k l m n ö p q r s t u v w x y z æ œ ñ ë ä ö b d d é é é j g g è y y q h l l l k l u m j n ñ ö è i j l r r s s f u ä ä y z z j l
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y y q h l l l k l u m j n ñ e è i j l r r s s f u ä ä y z z j l
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y y q h l l l k l u m j n ñ e è i j l r r s s f u ä ä y z z j l
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y y q h l l l k l u m j n ñ e è i j l r r s s f u ä ä y z z j l
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y y q h l l l k l u m j n ñ e è i j l r r s s f u ä ä y z z j l
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y y q h l l l k l u m j n ñ e è i j l r r s s f u ä ä y z z j l
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y y q h l l l k l u m j n ñ e è i j l r r s s f u ä ä y z z j l
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y y q h l l l k l u m j n ñ e è i j l r r s s f u ä ä y z z j l
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y y q h l l l k l u m j n ñ e è i j l r r s s f u ä ä y z z j l
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y y q h l l l k l u m j n ñ e è i j l r r s s f u ä ä y z z j l
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y y q h l l l k l u m j n ñ e è i j l r r s s f u ä ä y z z j l

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

Base and mark shaping – Complete

These grids display base and mark rendering in XeLaTeX documents in Libertinus serif regular, italic, semibold and semibold italic. The grids use more complete sets of Latin and IPA bases and above and below marks.

Blue indicates that XeLaTeX uses a precomposed Unicode character glyph.

Black indicates that the mark is positioned using GPOS anchors.

Red indicates HarfBuzz fallback shaping: no precomposed glyph and no GPOS anchors are used.

Light gray indicates that either the base or the mark glyph is missing from the font. In this case only the base glyph is shown.

Regular, Latin, Above

[illegible]

Regular patched, Latin, Above

[illegible]

Italic, Latin, Above

[illegible]

Italic patched, Latin, Above

[illegible]

Semibold, Latin, Above

[illegible]

Semibold patched, Latin, Above

[illegible]

[illegible]

[illegible]

Regular, Latin, Below

[illegible]

Regular patched, Latin, Below

[illegible]

Italic, Latin, Below

[illegible]

Italic patched, Latin, Below

[illegible]

Semibold, Latin, Below

[illegible]

Semibold patched, Latin, Below

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

Italic, IPA, Below

[illegible]

[illegible]

Semibold, IPA, Below

[illegible]

Semibold patched, IPA, Below

[illegible]

[illegible]

[illegible]

Capital alternates

These grids display capitals with alternate marks.

Regular, Capitals, Marks with capital alternates

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Ç È É Ê Ë Ì Í Î Ï Ñ Ò Ó Ô Õ Ö × Ø Ù Ú Û Ü Ý Þ ß à á â ã ä å æ ç è é ê ë ì í î ï ñ ò ó ô õ ö ÷ ø ù ú û ü ý þ ÿ

Regular patched, Capitals, Marks with capital alternates

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ç È É Ê Ë Ì Í Î Ï Ñ Ò Ó Ô Õ Ö × Ø Ù Ú Û Ü Ý Þ ß à á â ã ä å æ ç è é ê ë ì í î ï ñ ò ó ô õ ö ÷ ø ù ú û ü ý þ ß

Italic, Capitals, Marks with capital alternates

À Á Â Ã Ä Å Æ Ç È É Ê Ë Ì Í Î Ï Ñ Ò Ó Ô Õ Ö × Ø Ù Ú Û Ü Ý Þ ß à á â ã
ä å æ ç è é ê ë ì í î ï ñ ò ó ô õ ö ÷ ø ù ú û ü ý þ ß à á â ã
ä å æ ç è é ê ë ì í î ï ñ ò ó ô õ ö ÷ ø ù ú û ü ý þ ß à á â ã
ä å æ ç è é ê ë ì í î ï ñ ò ó ô õ ö ÷ ø ù ú û ü ý þ ß à á â ã
ä å æ ç è é ê ë ì í î ï ñ ò ó ô õ ö ÷ ø ù ú û ü ý þ ß à á â ã
ä å æ ç è é ê ë ì í î ï ñ ò ó ô õ ö ÷ ø ù ú û ü ý þ ß à á â ã
ä å æ ç è é ê ë ì í î ï ñ ò ó ô õ ö ÷ ø ù ú û ü ý þ ß à á â ã
ä å æ ç è é ê ë ì í î ï ñ ò ó ô õ ö ÷ ø ù ú û ü ý þ ß à á â ã
ä å æ ç è é ê ë ì í î ï ñ ò ó ô õ ö ÷ ø ù ú û ü ý þ ß à á â ã
ä å æ ç è é ê ë ì í î ï ñ ò ó ô õ ö ÷ ø ù ú û ü ý þ ß à á â ã
ä å æ ç è é ê ë ì í î ï ñ ò ó ô õ ö ÷ ø ù ú û ü ý þ ß à á â ã

Italic patched, Capitals, Marks with capital alternates

ĀBCDĒFGHĪJKLMNŌPQRSTŪVẀXỲZǾǾθΣUUYZEHIRSTVZÆĪŲ
 ĀBCĎĒFGHĪJKLMŃŌPǪŔŜTŪVẀXỲZǾǾθΣUUYZEHIRSTVZÆĪŲ
 ĀBCĎĒFGĤĪJKLMNŌPǪŔŜTŪVẀXỲZǾǾθΣUUYZEHIRSTVZĀĒĪŲ
 ĀBCDĒFGĤHĪJKLMNŌPQRSTŪVWXYZǾǾθΣUUYZEHIRSTVZĀĒĪŲ
 ĀBCĎĒFGĤHĪJKLMŃŌPǪŔŜTŪVẀXỲZǾǾθΣUŲŲZEHĪŔŜTVZÆĪŲ
 ĀBĈĎĒFGĤHĪJKLMŃŌPǪŔŜTŪVẀXỲZǾǾθΣŲŲŲZEHĪŔŜTVZÆĪŲ
 ĀBCĎĒFGHĪJKLMŃŌPǪŔŜTŪVẀXỲZǾǾθΣŲŲŲZEHĪŔŜTVZÆĪŲ
 ABCDEFGHIJKLMNŌPǪŔŜTŪVWXYZǾǾθΣUUYZEHIRSTVZÆĪŲ
 ĀBĈĎĒFGĤHĪJKLMŃŌPǪŔŜTŪVWXYZǾǾθΣUUYZEHIRSTVZÆĪŲ
 ĀBCDĒFGHĪJKLMNŌPǪŔŜTŪVWXYZǾǾθΣŲŲŲZEHĪŔŜTVZÆĪŲ
 ĀBCDĒFGHĪJKLMNŌPǪŔŜTŪVWXYZǾǾθΣUUYZEHIRSTVZÆĪŲ

Semibold, Capitals, Marks with capital alternates

[illegible]

Semibold patched, Capitals, Marks with capital alternates

À B C D È F G H Ì J K L M N Ò P Q R S T Û V W X Y Z Ɔ Ǝ Ë Ì Š Ů Ű Ŷ Ž È Ħ Ĭ Ŗ Š Ţ Ÿ Z Å Æ Ĩ Ñ Ò Û
Á Ā Ć Đ Ê Ğ Ĥ Ĩ Ĵ Ķ Ĺ Ń Ó Ĥ Ĵ Š Ţ Ů Ű Ŷ W X Y Z Ɔ Ǝ Ë Ì Š Ů Ű Ŷ Ž È Ħ Ĭ Ŗ Š Ţ Ÿ Z Å Æ Ĩ Ñ Ò Û
Ă Ą Ċ Ď ě Ĥ Ğ Ĥ Ĩ Ĵ Ķ Ĺ Ń Ó Ĥ Ĵ Š Ţ Ů Ű Ŷ W X Y Z Ɔ Ǝ Ë Ì Š Ů Ű Ŷ Ž È Ħ Ĭ Ŗ Š Ţ Ÿ Z Å Æ Ĩ Ñ Ò Û
Ą Ȧ Ć Đ Ê Ğ Ĥ Ĩ Ĵ Ķ Ĺ Ń Ó Ĥ Ĵ Š Ţ Ů Ű Ŷ W X Y Z Ɔ Ǝ Ë Ì Š Ů Ű Ŷ Ž È Ħ Ĭ Ŗ Š Ţ Ÿ Z Å Æ Ĩ Ñ Ò Û
Ȧ Ą Ċ Ď ě Ĥ Ğ Ĥ Ĩ Ĵ Ķ Ĺ Ń Ó Ĥ Ĵ Š Ţ Ů Ű Ŷ W X Y Z Ɔ Ǝ Ë Ì Š Ů Ű Ŷ Ž È Ħ Ĭ Ŗ Š Ţ Ÿ Z Å Æ Ĩ Ñ Ò Û
Ȧ Ą Ċ Ď ě Ĥ Ğ Ĥ Ĩ Ĵ Ķ Ĺ Ń Ó Ĥ Ĵ Š Ţ Ů Ű Ŷ W X Y Z Ɔ Ǝ Ë Ì Š Ů Ű Ŷ Ž È Ħ Ĭ Ŗ Š Ţ Ÿ Z Å Æ Ĩ Ñ Ò Û
Ȧ Ą Ċ Ď ě Ĥ Ğ Ĥ Ĩ Ĵ Ķ Ĺ Ń Ó Ĥ Ĵ Š Ţ Ů Ű Ŷ W X Y Z Ɔ Ǝ Ë Ì Š Ů Ű Ŷ Ž È Ħ Ĭ Ŗ Š Ţ Ÿ Z Å Æ Ĩ Ñ Ò Û
Ȧ Ą Ċ Ď ě Ĥ Ğ Ĥ Ĩ Ĵ Ķ Ĺ Ń Ó Ĥ Ĵ Š Ţ Ů Ű Ŷ W X Y Z Ɔ Ǝ Ë Ì Š Ů Ű Ŷ Ž È Ħ Ĭ Ŗ Š Ţ Ÿ Z Å Æ Ĩ Ñ Ò Û
Ȧ Ą Ċ Ď ě Ĥ Ğ Ĥ Ĩ Ĵ Ķ Ĺ Ń Ó Ĥ Ĵ Š Ţ Ů Ű Ŷ W X Y Z Ɔ Ǝ Ë Ì Š Ů Ű Ŷ Ž È Ħ Ĭ Ŗ Š Ţ Ÿ Z Å Æ Ĩ Ñ Ò Û
Ȧ Ą Ċ Ď ě Ĥ Ğ Ĥ Ĩ Ĵ Ķ Ĺ Ń Ó Ĥ Ĵ Š Ţ Ů Ű Ŷ W X Y Z Ɔ Ǝ Ë Ì Š Ů Ű Ŷ Ž È Ħ Ĭ Ŗ Š Ţ Ÿ Z Å Æ Ĩ Ñ Ò Û

Semibold italic, Capitals, Marks with capital alternates

[illegible]

Semibold italic patched, Capitals, Marks with capital alternates

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Ç È É Ê Ë Ì Í Î Ï Ñ Ò Ó Ô Õ Ö × Ø Ù Ú Û Ü Ý Þ ß à á â ã ä å æ ç è é ê ë ì í î ï ñ ò ó ô õ ö ÷ ø ù ú û ü ý þ ÿ